



Operate and maintain a 4-WD vehicle in safe working condition

D2.TTO.CL4.14

Trainer Guide



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Project Base

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Competency Based Training (CBT) and assessment – An introduction for trainers

Competency

Competency refers to the ability to perform particular tasks and duties to the standard of performance expected in the workplace.

Competency requires the application of specified knowledge, skills and attitudes relevant to effective participation, consistently over time and in the workplace environment.

The essential skills and knowledge are either identified separately or combined.

Knowledge identifies what a person needs to know to perform the work in an informed and effective manner.

Skills describe the application of knowledge to situations where understanding is converted into a workplace outcome.

Attitude describes the founding reasons behind the need for certain knowledge or why skills are performed in a specified manner.

Competency covers all aspects of workplace performance and involves:

- Performing individual tasks
- Managing a range of different tasks
- Responding to contingencies or breakdowns
- Dealing with the responsibilities of the workplace
- Working with others.

Unit of competency

Like with any training qualification or program, a range of subject topics are identified that focus on the ability in a certain work area, responsibility or function.

Each manual focuses on a specific unit of competency that applies in the hospitality workplace.

In this manual a unit of competency is identified as a 'unit'.

Each unit of competency identifies a discrete workplace requirement and includes:

- Knowledge and skills that underpin competency
- Language, literacy and numeracy
- Occupational safety and health requirements.

Each unit of competency must be adhered to in training and assessment to ensure consistency of outcomes.

Element of competency

An element of competency describes the essential outcomes within a unit of competency.

The elements of competency are the basic building blocks of the unit of competency. They describe in terms of outcomes the significant functions and tasks that make up the competency.

In this manual elements of competency are identified as an 'element'.

Performance criteria

Performance criteria indicate the standard of performance that is required to demonstrate achievement within an element of competency. The standards reflect identified industry skill needs.

Performance criteria will be made up of certain specified skills, knowledge and attitudes.

Learning

For the purpose of this manual learning incorporates two key activities:

- Training
- Assessment.

Both of these activities will be discussed in detail in this introduction.

Today training and assessment can be delivered in a variety of ways. It may be provided to participants:

- On-the-job – in the workplace
- Off-the-job – at an educational institution or dedicated training environment
- As a combination of these two options.

No longer is it necessary for learners to be absent from the workplace for long periods of time in order to obtain recognised and accredited qualifications.

Learning approaches

This manual will identify two avenues to facilitate learning:

Competency Based Training (CBT)

This is the strategy of developing a participant's competency.

Educational institutions utilise a range of training strategies to ensure that participants are able to gain the knowledge and skills required for successful:

- Completion of the training program or qualification
- Implementation in the workplace.

The strategies selected should be chosen based on suitability and the learning styles of participants.

Competency Based Assessment (CBA)

This is the strategy of assessing competency of a participant.

Educational institutions utilise a range of assessment strategies to ensure that participants are assessed in a manner that demonstrates validity, fairness, reliability, flexibility and fairness of assessment processes.

Flexibility in learning

It is important to note that flexibility in training and assessment strategies is required to meet the needs of participants who may have learning difficulties. The strategies used will vary, taking into account the needs of individual participants with learning difficulties. However they will be applied in a manner which does not discriminate against the participant or the participant body as a whole.

Catering for participant diversity

Participants have diverse backgrounds, needs and interests. When planning training and assessment activities to cater for individual differences, trainers and assessors should:

- Consider individuals' experiences, learning styles and interests
- Develop questions and activities that are aimed at different levels of ability
- Modify the expectations for some participants
- Provide opportunities for a variety of forms of participation, such as individual, pair and small group activities
- Assess participants based on individual progress and outcomes.

The diversity among participants also provides a good reason for building up a learning community in which participants support each other's learning.

Participant centred learning

This involves taking into account structuring training and assessment that:

- *Builds on strengths* – Training environments need to demonstrate the many positive features of local participants (such as the attribution of academic success to effort, and the social nature of achievement motivation) and of their trainers (such as a strong emphasis on subject disciplines and moral responsibility). These strengths and uniqueness of local participants and trainers should be acknowledged and treasured
- *Acknowledges prior knowledge and experience* – The learning activities should be planned with participants' prior knowledge and experience in mind
- *Understands learning objectives* – Each learning activity should have clear learning objectives and participants should be informed of them at the outset. Trainers should also be clear about the purpose of assignments and explain their significance to participants
- *Teaches for understanding* – The pedagogies chosen should aim at enabling participants to act and think flexibly with what they know
- *Teaches for independent learning* – Generic skills and reflection should be nurtured through learning activities in appropriate contexts of the curriculum. Participants should be encouraged to take responsibility for their own learning

- *Enhances motivation* – Learning is most effective when participants are motivated. Various strategies should be used to arouse the interest of participants
- *Makes effective use of resources* – A variety of teaching resources can be employed as tools for learning
- *Maximises engagement* – In conducting learning activities, it is important for the minds of participants to be actively engaged
- *Aligns assessment with learning and teaching* – Feedback and assessment should be an integral part of learning and teaching
- *Caters for learner diversity* – Trainers should be aware that participants have different characteristics and strengths and try to nurture these rather than impose a standard set of expectations.

Active learning

The goal of nurturing independent learning in participants does not imply that they always have to work in isolation or solely in a classroom. On the contrary, the construction of knowledge in tourism and hospitality studies can often best be carried out in collaboration with others in the field. Sharing experiences, insights and views on issues of common concern, and working together to collect information through conducting investigative studies in the field (active learning) can contribute a lot to their eventual success.

Active learning has an important part to play in fostering a sense of community in the class. First, to operate successfully, a learning community requires an ethos of acceptance and a sense of trust among participants, and between them and their trainers. Trainers can help to foster acceptance and trust through encouragement and personal example, and by allowing participants to take risks as they explore and articulate their views, however immature these may appear to be. Participants also come to realise that their classmates (and their trainers) are partners in learning and solving.

Trainers can also encourage cooperative learning by designing appropriate group learning tasks, which include, for example, collecting background information, conducting small-scale surveys, or producing media presentations on certain issues and themes. Participants need to be reminded that, while they should work towards successful completion of the field tasks, developing positive peer relationships in the process is an important objective of all group work.

Competency Based Training (CBT)

Principle of Competency Based Training

Competency based training is aimed at developing the knowledge, skills and attitudes of participants, through a variety of training tools.

Training strategies

The aims of this curriculum are to enable participants to:

- Undertake a variety of subject courses that are relevant to industry in the current environment
- Learn current industry skills, information and trends relevant to industry
- Learn through a range of practical and theoretical approaches
- Be able to identify, explore and solve issues in a productive manner

- Be able to become confident, equipped and flexible managers of the future
- Be 'job ready' and a valuable employee in the industry upon graduation of any qualification level.

To ensure participants are able to gain the knowledge and skills required to meet competency in each unit of competency in the qualification, a range of training delivery modes are used.

Types of training

In choosing learning and teaching strategies, trainers should take into account the practical, complex and multi-disciplinary nature of the subject area, as well as their participant's prior knowledge, learning styles and abilities.

Training outcomes can be attained by utilising one or more delivery methods:

Lecture/tutorial

This is a common method of training involving transfer of information from the trainer to the participants. It is an effective approach to introduce new concepts or information to the learners and also to build upon the existing knowledge. The listener is expected to reflect on the subject and seek clarifications on the doubts.

Demonstration

Demonstration is a very effective training method that involves a trainer showing a participant how to perform a task or activity. Through a visual demonstration, trainers may also explain reasoning behind certain actions or provide supplementary information to help facilitate understanding.

Group discussions

Brainstorming in which all the members in a group express their ideas, views and opinions on a given topic. It is a free flow and exchange of knowledge among the participants and the trainer. The discussion is carried out by the group on the basis of their own experience, perceptions and values. This will facilitate acquiring new knowledge. When everybody is expected to participate in the group discussion, even the introverted persons will also get stimulated and try to articulate their feelings.

The ideas that emerge in the discussions should be noted down and presentations are to be made by the groups. Sometimes consensus needs to be arrived at on a given topic. Group discussions are to be held under the moderation of a leader guided by the trainer. Group discussion technique triggers thinking process, encourages interactions and enhances communication skills.

Role play

This is a common and very effective method of bringing into the classroom real life situations, which may not otherwise be possible. Participants are made to enact a particular role so as to give a real feel of the roles they may be called upon to play. This enables participants to understand the behaviour of others as well as their own emotions and feelings. The instructor must brief the role players on what is expected of them. The role player may either be given a ready-made script, which they can memorize and enact, or they may be required to develop their own scripts around a given situation. This technique is extremely useful in understanding creative selling techniques and human relations. It can be entertaining and energizing and it helps the reserved and less literate to express their feelings.

Simulation games

When trainees need to become aware of something that they have not been conscious of, simulations can be a useful mechanism. Simulation games are a method based on "here and now" experience shared by all the participants. The games focus on the participation of the trainees and their willingness to share their ideas with others. A "near real life" situation is created providing an opportunity to which they apply themselves by adopting certain behaviour. They then experience the impact of their behaviour on the situation. It is carried out to generate responses and reactions based on the real feelings of the participants, which are subsequently analysed by the trainer.

While use of simulation games can result in very effective learning, it needs considerable trainer competence to analyse the situations.

Individual /group exercises

Exercises are often introduced to find out how much the participant has assimilated. This method involves imparting instructions to participants on a particular subject through use of written exercises. In the group exercises, the entire class is divided into small groups, and members are asked to collaborate to arrive at a consensus or solution to a problem.

Case study

This is a training method that enables the trainer and the participant to experience a real life situation. It may be on account of events in the past or situations in the present, in which there may be one or more problems to be solved and decisions to be taken. The basic objective of a case study is to help participants diagnose, analyse and/or solve a particular problem and to make them internalize the critical inputs delivered in the training. Questions are generally given at the end of the case study to direct the participants and to stimulate their thinking towards possible solutions. Studies may be presented in written or verbal form.

Field visit

This involves a carefully planned visit or tour to a place of learning or interest. The idea is to give first-hand knowledge by personal observation of field situations, and to relate theory with practice. The emphasis is on observing, exploring, asking questions and understanding. The trainer should remember to brief the participants about what they should observe and about the customs and norms that need to be respected.

Group presentation

The participants are asked to work in groups and produce the results and findings of their group work to the members of another sub-group. By this method participants get a good picture of each other's views and perceptions on the topic and they are able to compare them with their own point of view. The pooling and sharing of findings enriches the discussion and learning process.

Practice sessions

This method is of paramount importance for skills training. Participants are provided with an opportunity to practice in a controlled situation what they have learnt. It could be real life or through a make-believe situation.

Games

This is a group process and includes those methods that involve usually fun-based activity, aimed at conveying feelings and experiences, which are everyday in nature, and applying them within the game being played. A game has set rules and regulations, and may or may not include a competitive element. After the game is played, it is essential that the participants be debriefed and their lessons and experiences consolidated by the trainer.

Research

Trainers may require learners to undertake research activities, including online research, to gather information or further understanding about a specific subject area.

Competency Based Assessment (CBA)

Principle of Competency Based Assessment

Competency based assessment is aimed at compiling a list of evidence that shows that a person is competent in a particular unit of competency.

Competencies are gained through a multitude of ways including:

- Training and development programs
- Formal education
- Life experience
- Apprenticeships
- On-the-job experience
- Self-help programs.

All of these together contribute to job competence in a person. Ultimately, assessors and participants work together, through the 'collection of evidence' in determining overall competence.

This evidence can be collected:

- Using different formats
- Using different people
- Collected over a period of time.

The assessor who is ideally someone with considerable experience in the area being assessed, reviews the evidence and verifies the person as being competent or not.

Flexibility in assessment

Whilst allocated assessment tools have been identified for this subject, all attempts are made to determine competency and suitable alternate assessment tools may be used, according to the requirements of the participant.

The assessment needs to be equitable for all participants, taking into account their cultural and linguistic needs.

Competency must be proven regardless of:

- Language
- Delivery Method
- Assessment Method.

Assessment objectives

The assessment tools used for subjects are designed to determine competency against the 'elements of competency' and their associated 'performance criteria'.

The assessment tools are used to identify sufficient:

- a) Knowledge, including underpinning knowledge
- b) Skills
- c) Attitudes

Assessment tools are activities that trainees are required to undertake to prove participant competency in this subject.

All assessments must be completed satisfactorily for participants to obtain competence in this subject. There are no exceptions to this requirement, however, it is possible that in some cases several assessment items may be combined and assessed together.

Types of assessment

Allocated Assessment Tools

There are a number of assessment tools that are used to determine competency in this subject:

- Work projects
- Written questions
- Oral questions
- Third Party Report
- Observation Checklist.

Instructions on how assessors should conduct these assessment methods are explained in the Assessment Manuals.

Alternative assessment tools

Whilst this subject has identified assessment tools, as indicated above, this does not restrict the assessor from using different assessment methods to measure the competency of a participant.

Evidence is simply proof that the assessor gathers to show participants can actually do what they are required to do.

Whilst there is a distinct requirement for participants to demonstrate competency, there are many and diverse sources of evidence available to the assessor.

Ongoing performance at work, as verified by a supervisor or physical evidence, can count towards assessment. Additionally, the assessor can talk to customers or work colleagues to gather evidence about performance.

A range of assessment methods to assess competency include:

- Practical demonstrations
- Practical demonstrations in simulated work conditions
- Problem solving
- Portfolios of evidence
- Critical incident reports
- Journals
- Oral presentations
- Interviews
- Videos
- Visuals: slides, audio tapes
- Case studies
- Log books
- Projects
- Role plays
- Group projects
- Group discussions
- Examinations.

Recognition of Prior Learning

Recognition of Prior Learning is the process that gives current industry professionals who do not have a formal qualification, the opportunity to benchmark their extensive skills and experience against the standards set out in each unit of competency/subject.

Also known as a Skills Recognition Audit (SRA), this process is a learning and assessment pathway which encompasses:

- Recognition of Current Competencies (RCC)
- Skills auditing
- Gap analysis and training
- Credit transfer.

Assessing competency

As mentioned, assessment is the process of identifying a participant's current knowledge, skills and attitudes sets against all elements of competency within a unit of competency. Traditionally in education, grades or marks were given to participants, dependent on how many questions the participant successfully answered in an assessment tool.

Competency based assessment does not award grades, but simply identifies if the participant has the knowledge, skills and attitudes to undertake the required task to the specified standard.

Therefore, when assessing competency, an assessor has two possible results that can be awarded:

- Pass Competent (PC)
- Not Yet Competent (NYC).

Pass Competent (PC)

If the participant is able to successfully answer or demonstrate what is required, to the expected standards of the performance criteria, they will be deemed as 'Pass Competent' (PC).

The assessor will award a 'Pass Competent' (PC) if they feel the participant has the necessary knowledge, skills and attitudes in all assessment tasks for a unit.

Not Yet Competent' (NYC)

If the participant is unable to answer or demonstrate competency to the desired standard, they will be deemed to be 'Not Yet Competent' (NYC).

This does not mean the participant will need to complete all the assessment tasks again. The focus will be on the specific assessment tasks that were not performed to the expected standards.

The participant may be required to:

- a) Undertake further training or instruction
- b) Undertake the assessment task again until they are deemed to be 'Pass Competent'.

Competency standard

UNIT TITLE: OPERATE AND MAINTAIN A 4-WD VEHICLE IN SAFE WORKING CONDITION		NOMINAL HOURS: 60
UNIT NUMBER: D2.TTO.CL4.14		
UNIT DESCRIPTOR: This unit deals with skills and knowledge required to operate a <i>4-wheel drive</i> vehicle in a range of situations and provide for its basic maintenance while on tour		
ELEMENTS AND PERFORMANCE CRITERIA	UNIT VARIABLE AND ASSESSMENT GUIDE	
Element 1: Operate a 4-wheel drive vehicle 1.1 Engage 4-wheel drive correctly and when appropriate 1.2 Drive 4-wheel drive vehicle in a variety of <i>on-road conditions</i> and <i>off-road conditions</i> 1.3 Provide <i>safety and explanatory advice</i> to passengers 1.4 Ensure <i>minimal environmental impact</i> when driving a 4-wheel drive vehicle 1.5 Comply with road safety and road use rules and regulations Element 2: Recover a 4-wheel drive vehicle 2.1 Use a variety of <i>techniques to recover 4-wheel drive vehicle</i> from bogged or trapped situations 2.2 <i>Optimise safety for passengers and protection of property</i> 2.3 Inspect vehicle after recovery and <i>perform necessary maintenance</i> and repairs before proceeding	Unit Variables The Unit Variables provide advice to interpret the scope and context of this unit of competence, allowing for differences between enterprises and workplaces. It relates to the unit as a whole and facilitates holistic assessment. This unit applies to staff who operate a 4-wheel drive vehicle in a range of situations and provide basic maintenance while on tour, within the labour divisions of the hotel and travel industries and may include: 1. Tour Operation Engage 4-wheel drive refers to: • Selecting and engaging the 4-wheel drive according to manufacturer's instructions, including internal and external engagement • Selecting high or low range, or high or low model, depending on torque required • Selecting appropriate gear • Slowing or stopping to enable 4-wheel drive engagement, depending on vehicle type • Refraining from engaging optional 4-wheel drive on dry, hard, flat and firm surfaces • Protecting the safety of passengers and vehicle when engaging 4-wheel drive.	

Element 3: Provide basic maintenance for a 4-wheel drive vehicle 3.1 <i>Identify minor faults</i> 3.2 <i>Determine cause of faults</i> 3.3 <i>Conduct visual inspection of and operational tests on vehicle</i> 3.4 <i>Effect necessary repairs</i> 3.5 <i>Effect necessary preventative maintenance</i> Element 4: Report on 4-wheel drive operations 4.1 <i>Complete documentation to record services and repair undertaken</i> 4.2 <i>Notify management of issues that require external attention</i>	4-wheel drive vehicle may be: • Diesel or petrol powered • Made to manufacturer's instructions • Specially configured for touring conditions • Small capacity vehicles • Multi-seat units • Large or small coaches • Partial/part-time or permanent 4-wheel drive. <i>On-road conditions refer to sealed roads and may include:</i> • Slippery conditions, including rain and ice • Steep terrain • Snow. <i>Off-road conditions refer to terrain that is not sealed and may include:</i> • Steep inclines, including ascending and descending • Very soft ground • Rocky terrain • Sand • Water crossings • Mud. Safety and explanatory advice may be related to: • Advising the use of seat belts • Advising passengers to stay inside vehicle, including arms, hands and heads • Explaining how 4-wheel drive transmission works
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	• Stowing cargo appropriately, including re-packing cargo for off-road 4-wheel drive travel to attain a low centre of gravity • Taking pressure out of tyres. Ensure minimal environmental impact should include: • Adhering to designated tracks and routes in sensitive areas, parks or other prescribed areas • Assessing quality of tracks or intended routes before travelling on them • Minimising the effects of noise • Ensuring all required permits (including payment of fees), authorities and advice have been obtained and understood • Abiding by local prohibitions regarding entry by vehicles • Being vigilant to avoid damaging habitats, property, signs and other permanent and/or natural assets • Ensuring passengers do not litter the route and areas travelled • Ensuring passengers do not remove flora and/or fauna from the area • Not creating new tracks • Using passing bays, where provided • Leaving gates as they are found • Removing obstructions on tracks as opposed to driving round them • Not using trees as winch points if a recovery is required, or using tree protectors • Allowing animals to have right of way • Driving on beaches only on firm sand below high water marks.
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	<i>Techniques to recover 4-wheel drive vehicle</i> should relate to a variety of situations such as recovery from sand, mud, water and rocks, and may include: • Deflation of tyres before attempting to drive out • Selecting low range/ratio and low gear • Towing by another vehicle, including use of shackles, drag chains and snatch straps • Winching, including use of vehicle winch (electric, PTO, hydraulic and capstan), hand-operated winch, pulley blocks, winch extension straps and tree protectors • Using jacks, including kangaroo jack • Using under-vehicle inflatable bags • Using long handled shovel • Using portable tracks and/or sand ladders • Using front and rear recovery points. Optimise safety for passengers and protection of property may include: • Unloading items from the vehicle and removing to a safe and secure location • Disembarking passengers • Locating passengers at a safe distance (in a protected location) to avoid injury from flying parts that may occur during recovery. <i>Perform necessary maintenance</i> will most commonly apply to recoveries from water immersion and may include: • Checking no water in air intake • Remove spark plugs, or injectors and turn engine over to remove water • Replacing plugs, or injectors • Applying water displacement products to dry out high and low tension electrical circuits, as appropriate • Removing visible debris
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	• Re-aligning panel damage to enable free movement of wheels and steering • Making note to have differential and transmission checked and/or serviced at nearest opportunity. Identify minor faults may include: • Talking with co-drivers to identify type and nature of fault, including symptoms and indicators, frequency/conditions under which fault occurs/presents • Test driving vehicle to diagnose fault • Running engine of stationary vehicle to diagnose fault. Determine cause of faults may include: • Referring to service and operator manuals • Seeking advice from manufacturer, including over the phone assistance and radio communication • Eliminating factors from the scenario that is causing the fault on a sequential and structured basis while continuing ongoing testing • Referring to personal experience • Loading vehicle onto jacks to facilitate inspections • Use of applicable sensory appraisal, including sight, feel, smell and hearing • Monitoring on-board gauges and instruments. Conduct visual inspection of and operational tests on vehicle may include: • Checking for leaks • Checking condition of components, including hoses, belts, tyres, filters, radiator • Checking operation of indicators and lights • Checking engagement point of brake pedal and clutch pedal • Checking parking brake operation • Checking for noise and vehicle emissions • Inspecting battery.
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	Necessary repairs may include: • Replacing, adjusting or servicing items identified as unserviceable as a result of visual inspection and operational tests • Topping up and/or replacing fluids • Repairing punctures • Bleeding brakes • Bleeding a diesel engine • Replacing plugs. <i>Necessary preventative maintenance</i> will depend on the conditions in which the vehicle is being operated and may include: • Undertaking scheduled service tasks as recommended by the manufacturer • Undertaking specified additional service tasks as required by the host enterprise • Replacing component parts • Draining and replacing fluids, including topping up of fluids • Preparing the vehicle for conditions that are anticipated, including water and sand. <i>Documentation</i> may include: • On-board service records and logs. Notify management may include: • Verbal notification, including face-to-face, over the telephone, or via radio communication • Completing maintenance requisition • Discussing the problem with specialists.
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Assessment Guide

The following skills and knowledge must be assessed as part of this unit:

- The enterprise's policies and procedures in regard to the operation and maintenance of 4-wheel drive vehicles
- Principles of safe road use including on-road and off-road applications
- Knowledge of host country legislated and other requirements in relation to the service and maintenance of commercial passenger vehicles
- Principles of diagnosing vehicle faults
- Ability to use basic repair and maintenance tools, equipment and techniques
- Ability to apply knowledge of standard and basic maintenance and repair procedures
- Knowledge of vehicle manufacturer's instructions, specifications and recommended procedures for maintenance, trouble-shooting and repairs
- Knowledge of local road conditions
- Evidence of holding current and appropriate driver's license for the vehicle that is being driven.

Linkages To Other Units

- Establish and maintain safe touring conditions
- Drive large tour buses or coaches
- Manage operational risk
- Maintain tourism vehicles in safe and clean operational condition
- Drive various types of service vehicles
- Operate tours in remote areas.

	Critical Aspects of Assessment Evidence of the following is essential: • Understanding of host enterprise policies and procedures in regard to the operation and maintenance of 4-wheel drive vehicles • Understanding of the road rules and regulations of the host country and/or the road rules and regulations in which the 4-wheel drive vehicles are being driven • Proof of holding a current and valid driver's license to suit the 4-wheel drive vehicle • Demonstrated ability to perform required preventative maintenance and operational checks, as indicated on a supplied checklist, on a designated 4-wheel drive vehicle and provide written or verbal report on the condition of that vehicle • Demonstrated ability to diagnose at least three simulated, minor and unidentified faults on a nominated 4-wheel drive vehicle • Demonstrated ability to perform nominated and effective minor routine repairs to a designated 4-wheel drive vehicle in order to return the vehicle to a safe and effective operating condition • Demonstrated ability to drive a nominated 4-wheel drive vehicle in prescribed conditions so as to display at least seven of the following skill sets: ▪ Driving in sand ▪ Driving in mud ▪ Driving across rough terrain including rocky terrain ▪ Driving up and down steep inclines ▪ Driving through water ▪ Starting the vehicle, including key start and application of choke and hand controls to govern speed/engine revolutions ▪ Steering the vehicle and manoeuvring the vehicle, including uphill, downhill and on level ground ▪ Using all vehicle controls, instruments and indicators, including foot and hand-operated controls
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- Positioning and parking the vehicle
- Stopping the vehicle, including uphill, downhill and on level ground.

Context of Assessment

Assessment must ensure:

- Actual or simulated workplace application of vehicle maintenance and repair skills, and driving skills.

Resource Implications

Training and assessment must include the use of real vehicles, real tools and resources and a range of vehicles requiring a range of maintenance and repair provisions as well as the use of real vehicles in real or simulated driving conditions, situations and environments; and access to workplace standards, procedures, policies, guidelines, tools and equipment. Note: vehicle faults and problems may be simulated.

Assessment Methods

The following methods may be used to assess competency for this unit:

- Observation of practical candidate performance
- Portfolio of documents providing evidence of vehicle inspections, service, and repairs
- Oral and written questions
- Third party reports completed by a supervisor
- Project and assignment work.

	Key Competencies in this Unit <i>Level 1 = competence to undertake tasks effectively</i> <i>Level 2 = competence to manage tasks</i> <i>Level 3 = competence to use concepts for evaluating</i>		
	Key Competencies	Level	Examples
	Collecting, organising and analysing information	2	Check vehicles for faults and damage
	Communicating ideas and information	1	Convey safety information to passengers
	Planning and organising activities	1	Schedule vehicle serving; plan the best route to take given conditions that apply at the time
	Working with others and in teams	1	Drive safely and courteously with other road users; use passengers to assist in vehicle recovery
	Using mathematical ideas and techniques	1	Calculate quantities/volumes of fluids, air and other resources needed to service and repair vehicles
	Solving problems	2	Identify and diagnose faults
	Using technology	2	Drive vehicles and use on-board communication equipment; use servicing tools and equipment

Notes and PowerPoint slides

Slide

OPERATE AND MAINTAIN A 4-WD VEHICLE IN SAFE WORKING CONDITION

D2.TTO.CL4.14



Slide 1

Slide No	Trainer Notes
1.	Trainer welcomes students to class.

Slide

Operate and maintain a 4-WD vehicle in safe working condition

This Unit comprises four Elements:

- 1. Operate a 4-wheel drive vehicle
- 2. Recover a 4-wheel drive vehicle
- 3. Provide basic maintenance for a 4-wheel drive vehicle
- 4. Report on 4-wheel drive operations



Slide 2

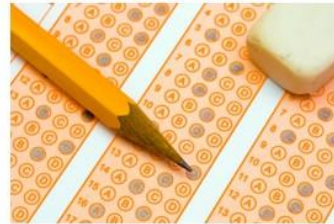
Slide No	Trainer Notes
2.	Trainer advises trainees this Unit comprises four Elements, as listed on the slide explaining: ● Each Element comprises a number of Performance Criteria which will be identified throughout the class and explained in detail ● Trainees can obtain more detail from their Trainee Manual ● At times the course presents advice and information about various protocols but where their workplace requirements differ to what is presented, the workplace practices and standards, as well as policies and procedures must be observed.

Slide

Assessment

Assessment for this unit may include:

- Oral questions
- Written questions
- Work projects
- Workplace observation of practical skills
- Practical exercises
- Formal report from employer/supervisor



Slide 3

Slide No	Trainer Notes
3.	Trainer advises trainees that assessment for this Unit may take several forms, all of which are aimed at verifying they have achieved competency for the Unit as required. Trainer indicates to trainees the methods of assessment that will be applied to them for this Unit.

Slide

Element 1 – operate a 4-wheel drive vehicle

Performance Criteria for this Element are:

- Engage 4-wheel drive correctly and when appropriate
- Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions
- Provide safety and explanatory advice to passengers
- Ensure minimal environmental impact when driving a 4-wheel drive vehicle
- Comply with road safety and road use rules and regulations



Slide 4

Slide No	Trainer Notes
4.	Trainer identifies for trainees the Performance Criteria for this Element, as listed on the slide. Class Activity – General Discussion Trainer leads a general class discussion by asking questions such as: • What 4WDs have you driven? • Where? • When? • In what conditions?

Slide

Engage 4-wheel drive correctly and when appropriate

Four-wheel drives may be:

- Diesel or petrol
- Standard models
- Built to specifications
- Configured to seat a range of passengers up to 20+



Slide 5

Slide No	Trainer Notes
5.	Trainer explains a wide variety of vehicles are used for four-wheel drive touring – they can include: • Diesel or petrol powered engines – the choice of fuel is very much an individual consideration with access to fuel for re-supply a primary consideration ▪ Personal preference is also a significant factor with many believing 'diesel is best' while many others believe the opposite • Standard 'off-the-shelf' vehicles – which: ▪ Are produced according to manufacturers' specifications ▪ May be bought with a wide range of options fitted from the factory ▪ Can be optioned with after-market options to suit individual need • Vehicles produced to Tour Operator specifications – which means they are specially configured ▪ For identified touring conditions ▪ To accommodate the experience the Operator has built up over the years

- A range of configurations – including:
 - Small capacity vehicles (cars, SUVs, utilities, wagons, motor homes) accommodating four to five people
 - Multi-seat units (people movers, troop carriers) with space for seven to 12 people
 - Large or small coaches which can seat 20 plus.

Classroom Activity – Internet research

Trainer supervises internet research on relevant websites such as:

http://www.outbackcrossing.com.au/FourWheelDrive/Petrol_or_Diesel_4WD.shtml - Petrol or diesel 4WD?

<http://www.4x4australia.com.au/drive/road-tests/0808/petrol-versus-diesel/> - Petrol versus diesel.

Slide

Engage 4-wheel drive correctly and when appropriate

Partial 4WD:

- Is two-wheel drive when on sealed roads/under normal conditions
- Becomes 4WD when the other two wheels/hubs are 'locked in'
- Generally give High and Low range options



Slide 6

Slide No	Trainer Notes
6.	Trainer discusses partial 4WD: - It means the vehicle is usually a 2-wheel drive vehicle when travelling normal roads/highways but has the capacity to become a 4WD when 4WD is activated/selected - The two wheels which are not under power are known as free-wheeling hubs. High and Low range Part-time 4WDs generally give the driver the option of selecting 'High' or 'Low' range 4WD. Low range provides/generates more torque at the wheels through lower gear ratios which is useful for driving difficult terrain at low speeds. High range gives greater control and traction when driving quicker on surfaces such as compacted sand/beaches. 2WD A vehicle that is only powered by one set of wheels (either front or rear) is known as a two-wheel drive/2WD.

Slide

Engage 4-wheel drive correctly and when appropriate

Vehicles may also be:

- AWD = permanent 4WD where vehicle is always in four-wheel drive
- A4WD = where vehicle senses conditions and automatically selects/engages 4WD when necessary



Slide 7

Slide No	Trainer Notes
7.	Trainer describes other 4WD vehicle options: Permanent 4-wheel drive A permanent 4-wheel drive vehicle is normally in 4WD mode. There is therefore no need to engage 4WD because it is always selected. There can be the facility to de-select 4WD and engage 2WD. Permanent 4-wheel drives are also referred to all-wheel drive (AWD) vehicles. Automatic four-wheel drive This is a system where the vehicle senses different individual wheel speeds and engages 4WD automatically it detects a need to do so. These are known as A4WDs.

Slide

Engage 4-wheel drive correctly and when appropriate

Terminology:

- 4x4 = vehicle with 4 wheels and drive to all 4 wheels
- 4x2 = vehicle with 4 wheels and drive to only 2 wheels
- 6x6 = vehicle with 6 wheels and drive to all 6 wheels



Slide 8

Slide No	Trainer Notes
8.	Trainer explains terminology: 4x4 and 4x2 4x4 = a vehicle that has four wheels (the first 4) and has drive to all 4 wheels (the second 4) 4x2 = a conventional 2WD vehicle with 4 wheels and power only to two of them 6x6 = a vehicle that has six wheels and has drive to all 6 wheels.

Slide

Engage 4-wheel drive correctly and when appropriate

Differentials:

- Part of the drive train which transfers/converts power from the drive shaft to the wheels
- Gears in the diff allow wheels on opposite sides of the vehicle to turn at different speeds
- 4WDs can have 2 (front and rear) or 3 diffs (centre)

(Continued)



Slide 9

Slide No	Trainer Notes
9.	Trainer discusses differentials: Background The engine develops power which is transferred to a drive shaft. The drive shaft is connected to the axles using a differential which transfers the power in the drive shaft to axles, and on to the wheels. The gears in the differential allow the drive wheels on opposite sides of the vehicle to rotate at different speeds to accommodate optimum performance when turning corners or in slippery conditions. 4WD vehicles have (at least) two diffs – one for the front wheels and one for the rear wheels. Some makes and models also have a third diff (a centre diff) which enables the front and rear drive shafts to also turn at different speeds, as well as letting each wheel rotate/drive at different speeds. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: http://www.lcool.org/technical/diffs/diffs.html - Land Cruiser Owners Online: Diffs for Dummies https://www.youtube.com/watch?v=gIGvhvOhLHU – How a differential works and types of differentials

Classroom Activity (2) – Demonstration

Trainer shows students 4WD vehicles indicating location of diffs and showing how they operate.

Slide

Engage 4-wheel drive correctly and when appropriate

Differentials may also feature:

- Limited slip diffs
- Diff locking



Slide 10

Slide No	Trainer Notes
10.	Trainer presents further information on differentials: Limited Slip Diffs Some vehicles may be fitted with LSDs. A LSD allows more power from the drive shaft to be sent to the wheel with the <i>most</i> traction in a situation where one of the wheels is slipping/spinning. This means extra (but limited) drive can be gained from the vehicle via the wheel which is getting the most purchase on the road/track surface. Diff locking Diff locking is where the differential is locked so it drives both the axles (and hence both the wheels) with the same speed/power regardless of the road surface/condition. Depending on the make/model of the 4WD it can be possible to: Lock the front diff; Lock the rear diff; Lock the centre diff This facility is often a benefit in certain four-wheel driving conditions as it can give much better traction, drive, control and ability to get through sand, bogs and difficult terrain. Some vehicles (A4WDs) are fitted with automatic diff lockers meaning the vehicle senses when they are needed, and turns them on and off as required on the basis of information provided by sensors.

It is important to unlock manually applied diff locks when the difficult driving conditions are overcome and the vehicle returns to 'normal' use (so the wheels are able to rotate at different speeds as required when turning),

Classroom Activity (1) – Internet research

Trainer supervises internet research on relevant websites such as:

<https://www.youtube.com/watch?v=WeLm7wHvdxQ> – Understanding limited slip differentials

<https://www.youtube.com/watch?v=RcRBYpccP6g> – How diff locks work and when to use it.

Classroom Activity (2) – Demonstration

Trainer shows students 4WD vehicles demonstrating operation of:

- Limited slip diff
- Diff locking.

Slide

Engage 4-wheel drive correctly and when appropriate

Tour Operators use 4WDs because:

- Customers expect them
- Customers like the 4WD experience
- They give customers confidence and a feeling of safety

(Continued)



Slide 11

Slide No	Trainer Notes
11.	Trainer gives reasons for Tour Operators using 4WDs: • Their customers expect them to be using one • Customers like the often unique nature of an off-road 4WD experience, driving through water and mud • Four-wheel drives give customers confidence in the vehicle and the fact the tour will be a safe and trouble-free one.

Slide

Engage 4-wheel drive correctly and when appropriate

- They give greater ground clearance and ramp-over
- They allow better approach and departure angles
- They handle off-road conditions better



Slide 12

Slide No	Trainer Notes
12.	Trainer continues giving reasons for Tour Operators using 4WDs: • They give greater ground clearance and ramp-over angles than conventional vehicles • They allow better approach and departure angles when (for example) approaching river crossing and clearing jump-ups • They enable more dependable manoeuvrability of off-road, difficult and challenging conditions such as loose sand and mud – meaning a 4WD can take people places a conventional 2WD vehicle cannot go.

Slide

Engage 4-wheel drive correctly and when appropriate

Context for four-wheel driving:

- Engage 4WD before it is needed
- Momentum is important
- Tyre pressures should often be lowered
- Follow manufacturer's instructions
- There are limits to what 4WD can do
- Chains may be required



Slide 13

Slide No	Trainer Notes
13.	Trainer provides context relating to four-wheel driving: ● Four-wheel drive needs to be selected before it is needed (see below) – based on: ▪ Observation and analysis of the upcoming conditions of the route to be taken ▪ Personal experience ▪ Getting into trouble and then selecting 4WD is to be avoided – as: - It may be difficult/impossible to select 4WD when the vehicle is stuck - It may not resolve the situation ● Momentum is important – four-wheel drive <i>alone</i> is insufficient to get a vehicle through some conditions ● Tyre pressure is important in some situations – generally, lowering pressure in tyres will result in better traction through an increase in the footprint of the tread of the tyres on the surface ● Vehicles should be operated strictly in accordance with any specifications provided for by the manufacturer of the vehicle as stated in the Owner/User manual ● There are limits as to what a 4WD can do – they cannot 'do anything and go anywhere' ● Four-wheel drive may need to be supplemented by fitting chains to the tyres in some conditions such as snow or mud – to give added traction and control.

Classroom Activity – Guest Speaker

Trainer arranges for experienced driver of tour 4WDs to attend and:

- Discuss their experience and qualifications
- Explain why 4WDFs are used and how they are used
- Describe benefits of tour 4WDs
- Indicate limitations of vehicles.

Slide

Engage 4-wheel drive correctly and when appropriate

Four-wheel drive may need to be engaged:

- In muddy conditions
- In rough/rocky terrain
- In soft sand
- In snow and ice
- On steep slopes
- When crossing through water



Slide 14

Slide No	Trainer Notes
14.	Trainer indicates there can be a need to engage 4WD: • When tracks are muddy/boggy • If the terrain is rough and rocky • Where there is soft sand • In the event of snow and ice • Going up or down steep inclines • When crossing creeks/rivers.

Slide

Engage 4-wheel drive correctly and when appropriate

When engaging 4WD always:

- Follow manufacturer's instructions
- Set a comfortable driving position
- Practice
- Avoid 4WD where long, fast travel is intended



Slide 15

Slide No	Trainer Notes
15.	Trainer highlights engaging four-wheel drive will vary depending on the make and model of the vehicle but always: • Follow manufacturer instructions • Create a comfortable driving/sitting position enabling full operation of all foot and hand-operated controls (knobs and pedals, levers and switches) and clear line of sight around the vehicle through windscreens, windows and mirrors • Practice the procedure before there is a need to implement it in a real-life tour situation • Refrain from engaging optional 4-wheel drive on dry, hard, flat and firm surfaces – as this increases fuel consumption and wear and tear on parts.

Slide

Engage 4-wheel drive correctly and when appropriate

Indicative actions when engaging 4WD may require:

- Stop the vehicle
- Move gear shift to neutral
- Lock in hubs
- Move selector to High or Low range, or

(Continued)



Slide 16

Slide No	Trainer Notes
16.	Trainer presents indicative actions when engaging 4WD: • Stopping the vehicle – apart from vehicles which have 'on the fly/move' 4WD allowing engagement of four-wheel drive while travelling below a nominated speed • Moving gear shift to neutral • Locking in the hubs of the vehicle – by getting out of the vehicle and moving the locking hubs from the 'Free' position to the 'Lock' position • Moving 4WD selector to either High range or Low range (or Lock) – depending on conditions/torque required.

Slide

Engage 4-wheel drive correctly and when appropriate

- Push a designated button on the dash, or
- Move a selector knob
- Select the appropriate gear for the work/travel required



Slide 17

Slide No	Trainer Notes
17.	Trainer continues to present indicative actions when engaging 4WD: • Pushing a designated 4WD button on the dash • Moving a selector knob to the required position • Selecting appropriate gear. Classroom Activity – Demonstration and Practical Trainer demonstrates how to engage 4WD and gives students an opportunity to practice same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Golden Keys to successful four-wheel driving:

- Practice
- Experience



Slide 18

Slide No	Trainer Notes
18.	Trainer stresses the Golden Keys to successful four-wheel driving are the two related factors of: • Practice • Experience. It is critical drivers of 4WD tour vehicles gain competency in driving their vehicle in all expected on- and off-road conditions <i>before</i> they do so in a real-life touring environment where paying passengers/tour groups are involved.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Basic conditions:

- Off-road = terrain that is not sealed
- On-road = sealed roads/highways



Slide 19

Slide No	Trainer Notes
19.	Trainer describes basic road conditions: • Off-road conditions refer to terrain/ground that is not sealed or paved • On-road conditions refer to driving on sealed roads/highways.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Driving 4WDs in on-road conditions may include driving:

- In dry conditions/weather (H2)
- In the wet/rain (H4)
- In snow and ice (Low range)



Slide 20

Slide No	Trainer Notes
20.	Trainer explains driving 4WDs in on-road conditions may include driving: ● In dry conditions – where arguably most driving is done, and the standard approach is to: ▪ Use two-wheel drive/H2 only ▪ Leave extra distance between vehicles to allow for the extra braking distances required for heavier 4WD vehicles ▪ Take tight corners at a lower speed due to the relatively high centre of gravity for 4WDs ▪ Drive with headlights on at all times ● In wet conditions such as when there is rain and/or ice – where speed must be reduced and High range 4WD can be used ● In snow – where speed must be reduced, chains may need to be fitted and Low range may need to be selected.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Off-road 4WDing may involve:

- Steep inclines
- Sand
- Rocky terrain
- Water
- Soft ground
- Mud



Slide 21

Slide No	Trainer Notes
21.	Trainer explains operating a 4WD in off-road conditions can require/involve driving: • Steep inclines – including ascending and descending • Sand • Rocky terrain • Water crossings • Very soft ground • Mud.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Before driving off-road:

- Prepare the vehicle for expected conditions
- Undertake basic checks

(Continued)



Slide 22

Slide No	Trainer Notes
22.	Trainer gives pre-requisites before driving off-road: ● Ensure the vehicle has been properly prepared for the conditions which are known to exist or which are expected on the trip – this may include: ▪ Loading the vehicle with necessary recovery gear ▪ Fitting protection and bash plates to the underside of the vehicle ▪ Fitting appropriate tyres ▪ Having tyre deflators and pumps ▪ Having warning/sand flags ▪ Having chains ▪ Fitting a snorkel ▪ Fitting a viscous fan ▪ Fitting extended differential breathers ● Undertake basic checks – to confirm: ▪ The vehicle is fully operational – to the nest extent possible under the circumstances that apply at the time ▪ Safety and recovery gear is available and operational.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Inspect the route
- Be prepared to choose another route (or wait)

(Continued)



Slide 23

Slide No	Trainer Notes
23.	Trainer continues to give pre-requisites before driving off-road: • Inspect or consider the way ahead – to: ▪ Determine the route/path to be taken ▪ Identify potential obstacles/problems ▪ Make the final 'Go' or 'No go' decision • Always be prepared to, as required: ▪ Choose or find an alternate route ▪ Back-track and go through using another track ▪ Wait for poor conditions (high water levels, high flow rates of rivers, deep snow) to abate/improve before proceeding.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Make passenger safety a primary consideration
- Engage 4WD



Slide 24

Slide No	Trainer Notes
24.	Trainer continues to give pre-requisites before driving off-road: ● Consider passenger safety – which may involve: ▪ Communicating with them to advise them of what to expect ▪ Providing safety information ▪ Keeping them involved and included in the tour experience ▪ Taking controlled risks can be part of the excitement and experience of a 4WD adventure but they must never be undertaken where they pose an uncontrolled threat to people/passengers ● Engage four-wheel drive – as appropriate for the individual vehicle and the nature of the route and terrain to be driven.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Ascending steep inclines:

- Inspect the path/route
- Engage 4WD
- Engage diff locks
- Turn off A/C

(Continued)



Slide 25

Slide No	Trainer Notes
25.	Trainer describes procedures for ascending steep inclines: ● Inspect the path – looking for: ▪ A suitable route through/across the track which presents itself – taking into account, for example, ruts, holes, fallen trees, rocks ▪ Surface condition – in relation to whether the ground is wet or dry and whether it is hard underfoot or soft ▪ Approach angle – to ensure the vehicle can begin the climb and get its wheels onto the incline, and not get ‘beached’/stuck trying to make the climb ▪ General angle/severity of the climb ▪ Length of the incline – in terms of the distance to be covered to complete the climb ▪ Possible recovery points that can be used – if the vehicle becomes stuck ● Engage 4WD – which may require locking in hubs ● Engage diff locks – if fitted and if necessary ● Turn off air conditioner – to avoid inconsistent engine acceleration.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Begin climbing in a manner reflecting the slope and conditions
- Stay square on
- Work wheel side-to-side to get extra grip if needed
- Pause at the top



Slide 26

Slide No	Trainer Notes
26.	Trainer continues to describe procedures for ascending steep inclines: ● Begin the climb in a manner that reflects the findings of observation of the incline – which may require, as appropriate: ▪ Reversing and taking a run at the incline to gain momentum perhaps in H4 where the ground is solid, the length of the climb is short and the angle of incline is not too steep ▪ Selecting Low range first or second gear and driving slowly up the incline where it is a steep incline, ground is soft sand/or the distance to be covered is significant ● Stay square to the incline – climbing at an angle to the rise introduces the potential for the vehicle to roll ● Work the wheel a little from side-to-side if the wheels start spinning – in order to use the tread on the sides of the tyres to gain additional purchase/traction ● Pause at the top of the incline – in order to: ▪ Check the descent – if there is one ▪ Check for oncoming vehicles. Classroom Activity – Demonstration and Practical Trainer demonstrates how to climb steep incline in 4WD and gives students an opportunity to practice same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Stall recovery of steep ascent:

- Apply foot brake
- Apply handbrake
- Turn ignition off
- Depress clutch
- Select reverse gear

(Continued)



Slide 27

Slide No	Trainer Notes
27.	Trainer explains if the vehicle stalls when climbing a steep incline the procedure for a safe stall recovery is: • Apply foot brake as vehicle stalls • Apply handbrake • Turn ignition off • Depress clutch • Select reverse gear.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Release clutch
- Release handbrake slowly
- Turn on ignition and release brakes
- Start vehicle allowing it to reverse down the slope
- Apply brakes as necessary to control descent



Slide 28

Slide No	Trainer Notes
28.	Trainer continues to explain ascending stall recovery: ● Release clutch ● Release handbrake slowly ● Turn on ignition and release brakes ● Start vehicle allowing it to reverse down the slope ● Apply brakes as necessary to control descent. Classroom Activity – Demonstration and Practical Trainer demonstrates how to perform stall recovery when climbing steep incline in 4WD and gives students an opportunity to practice same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Descending steep inclines:

- Inspect the path/route
- Engage 4WD
- Engage diff locks
- Turn off A/C

(Continued)



Slide 29

Slide No	Trainer Notes
29.	Trainer describes procedures for descending steep inclines: ● Inspect the path – looking for: ▪ A suitable route through/across the track which presents itself – taking into account, for example, ruts, holes, fallen trees, rocks ▪ Surface condition – in relation to whether the ground is wet or dry and whether it is hard underfoot or soft ▪ Departure angle at the end of the incline – to ensure the vehicle can leave the incline and make it to level ground instead of being stuck half-way between the incline and the flat ground ▪ Length of the descent – in terms of the distance to be covered to complete the descent ▪ General angle/severity of the descent ▪ Possible run-off points (into banks, bushes, trees) – in case the descent becomes out-of-control ● Engage 4WD – which may require locking in hubs ● Engage diff locks – if fitted and if necessary ● Turn off air conditioner – to avoid inconsistent engine acceleration.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Begin descent to reflect conditions
- Select Low range, first gear
- Allow vehicle to creep forward
- Minimise braking – let engine do the braking
- Keep wheels straight/pointing forward
- Stay square on



Slide 30

Slide No	Trainer Notes
30.	Trainer continues to describe procedures for descending steep inclines: ● Begin travel in a manner that reflects the findings of observation of the descent – which may include: ● Selecting Low range first gear ● Allowing vehicle to creep forward under its own power without using the accelerator – letting the engine act as a brake ● Applying extra braking as required using either foot brake or hand brake – to keep a safe speed ▪ Minimising use of brake – allowing engine to do the braking ● Avoiding locking of wheels – the wheels must continue to rotate in order to retain control/steering ● Keeping wheels pointing forward ● Stay square on to the incline – to avoid potential for roll-over. Classroom Activity – Demonstration and Practical Trainer demonstrates how to descend steep incline in 4WD and gives students an opportunity to practice same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Stall recovery of steep descent:

- Turn ignition off
- Apply foot brake
- Engage handbrake
- Depress clutch
- Select low gear
- Release clutch

(Continued)



Slide 31

Slide No	Trainer Notes
31.	Trainer explains if the vehicle stalls when descending a steep incline the procedure for a safe stall recovery is: • Turn ignition off • Apply foot brake • Engage handbrake • Depress clutch • Select low gear • Release clutch. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://www.4x4australia.com.au/gear/1004/4x4-steep-slopes-tips/ - 4x4 steep slope tips http://4xoverland.com/how-do-4x4s-work/4x4-driving/ - How to drive a 4x4 off road [Steep slopes] https://www.youtube.com/watch?v=uTW3H4WBxnY – 4WD Hill Climb Comparo http://www.iventureclub.com.au/4x4-tips/article.html?k=Jem4UGra2DY – 4x4: Tip-Steep terrain

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Slowly release handbrake
- Turn ignition on
- Release brakes
- Start vehicle and allow to continue down the slope
- Apply brakes emulating ABS as necessary to control descent
- Steer into skids



Slide 32

Slide No	Trainer Notes
32.	Trainer continues to explain descending stall recovery: ● Slowly release handbrake ● Turn ignition on ● Release brakes ● Start vehicle and allow to continue down the slope ● Apply brakes emulating ABS as necessary to control descent ● Steer into skids. Classroom Activity – Demonstration and Practical Trainer demonstrates how to perform stall recovery when descending steep incline in 4WD and gives students an opportunity to practice same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Preparing for soft sand 4WD driving:

- Lower tyre pressure of vehicle
- Lower tyre pressure of trailer
- Use tyre pressure gauge
- Be prepared to lower pressure a number of times



Slide 33

Slide No	Trainer Notes
33.	Trainer talks about preparing for soft sand: ● Lower tyre pressure of 4WD vehicle – as a starting point to increase the grip area of the tread and help keep it on top of the sand instead of sinking (or biting) into it: ▪ Reduce by half standard on-road pressure ▪ Reduce more for softer surfaces ● Lower tyre pressure of trailer, where appropriate ● Use pressure gauge to determine pressures – so there is a known reference point: write the pressures down as they are obtained and trialled ● Be prepared to lower pressures further if first decrease appears insufficient – reduce by approx. two to four psi (15 – 30 kPa) each time. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: http://www.offroadaussie.com/2013/02/4wd-tips-and-tricks-tyre-pressure/ - 4WD tricks and tips: tyre pressure. Classroom Activity (2) – Demonstration and Practical Trainer shows students how to lower tyre pressures and provides opportunity for them to do the same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

When driving on hard sand:

- Lower tyre pressure slightly
- Engage 4WD, High range
- Decide on gear use with reference to conditions and vehicle
- Travel at reduced speed
- Do not contact salt water
- Re-inflate tyres when sand travel is completed



Slide 34

Slide No	Trainer Notes
34.	Trainer discusses driving on hard sand observing it is a bit like driving on sealed roads: • Reduce tyre pressure slightly – to increase surface area of tread that makes contact with the sand • Engage 4WD – which may require locking in hubs • Select High range • Drive off choosing gear/s appropriate to conditions, weight of vehicle, route being used and desired speed of travel • Travel at a reduced speed – that is, slower than on the open highway • Avoid contact with salt water wherever possible – to prevent potential of vehicle being damaged by rust: always wash any vehicle as soon as possible that has been contacted by salt water • Re-inflate tyres when sand driving is completed – if they have been tyre pressure was initially reduced.

Classroom Activity (1) – Internet research

Trainer supervises internet research on relevant websites such as:

<http://4xoverland.com/how-do-4x4s-work/4x4-driving/> - How to drive a 4x4 off road [Sand driving]

<https://www.youtube.com/watch?v=iT0Zph4h-L4> – ExpeditionOz 4WD Skills: Sand driving

Classroom Activity (2) – Demonstration and Practical

Trainer shows students how to drive on hard sand and provides opportunity for them to do the same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

When driving on soft sand:

- Lower tyre pressures
- Attach sand warning flags
- Check/ensure recovery gear is available
- Engage 4WD, Low range, 2nd gear

(Continued)



Slide 35

Slide No	Trainer Notes
35.	Trainer discusses driving on soft sand: ● Deflate tyres – to around 16 psi/110kPa or lower depending on sand depth/condition ● Attach sand/warning flags to the vehicle if driving in areas where there are hills/dunes – to help indicate location of the vehicle to others in the area ● Check recovery gear is available – to use should the vehicle become stuck ● Engage 4WD – which may require locking in hubs ● Select Low range – second gear is a good starting point. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=I2bA6aC4g50 – Soft sand driving https://www.youtube.com/watch?v=KHI_cMrvsrM – 4x4: How to drive on sand tracks

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Start slowly and drive slowly
- Avoid sudden turns
- Drive in established tracks
- Be ready for less feedback through the steering wheel

(Continued)



Slide 36

Slide No	Trainer Notes
36.	Trainer continues to discuss driving on soft sand: • Start slowly – to avoid wheel spin and digging in • Drive slowly but maintain momentum • Avoid sudden turns • Drive in established tracks – where the sand will be more compacted by the vehicles which have previously used that route • Be prepared for less feedback from the steering wheel – due to the depth of the sand. Classroom Activity – Internet research http://www.outbackcrossing.com.au/FourWheelDrive/How_To_Drive_On_Sand.shtml - How to drive on sand https://www.youtube.com/watch?v=Int_D9zKjiY – Tips for beginners: Driving on sand, dunes and beaches

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Approach dunes square on
- Stay alert for oncoming traffic over dunes
- Never assume 'down' side of dunes will be the same as the 'up' side
- Be prepared to back-up and repeat a failed attempt with more momentum
- Re-inflate tyres when sand travel is completed



Slide 37

Slide No	Trainer Notes
37.	Trainer continues to discuss driving on soft sand: ● Approach any dunes head on/square on if possible – to avoid potential for roll-over ● Stay alert for oncoming traffic over the top of dunes ● Never assume the drop off on the other side of a dune is the same as the climb up the dune – drop-offs on the other side can be substantially different due to wind action which may have created a sheer fall/drop away on the other side ● Use the brake gently – to avoid sand building up in front of the tyres ● Be prepared to reverse back over the tracks made by the vehicle if stuck – as these will be more compacted and may provide sufficient traction to develop the required traction and momentum to move forward ● Re-inflate tyres when sand driving is completed – if they have been tyre pressure was initially reduced. Classroom Activity – Demonstration and Practical Trainer shows students how to drive on soft sand and provides opportunity for them to do the same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

When there is a need to drive rocky terrain re-consider the route is possible because it poses risks in terms of:

- Uncomfortable ride for passengers
- Slow travel times
- Punctures
- Damage to the vehicle

(Continued)



Slide 38

Slide No	Trainer Notes
38.	Trainer provides base information for driving rocky terrain: ● In general, rocky terrain should be avoided wherever possible as it poses lots of risks in terms of: ▪ Uncomfortable ride for passengers ▪ Slow travel times – which may negatively impact itinerary ▪ Punctures ▪ Damage to the vehicle especially the chassis and sub-structure/under-side parts of the vehicle such as diffs and fuel tanks.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

When driving rocky terrain:

- Inspect the route
- Lower tyre pressures
- Engage 4WD and diff locks
- Turn off air conditioner
- Begin in a manner that reflects the inspection of the route

(Continued)



Slide 39

Slide No	Trainer Notes
39.	Trainer discusses driving on rocky terrain: ● Inspect the path – as described previously ● Reduce pressures in tyres ● Engage 4WD – which may require locking in hubs ● Engage diff locks – if fitted and if necessary ● Turn off air conditioner – to avoid inconsistent engine acceleration ● Begin in a manner that reflects the findings of observation – which may require, as appropriate: ▪ Selecting Low range first or second gear ▪ Driving slowly ('crawling') and smoothly to avoid wheel spin ▪ Intelligently choosing the path to take to avoid traps, rocks and beaching of the vehicle.

Classroom Activity – Internet research

Trainer supervises internet research on relevant websites such as:

<http://www.4-wheeling-in-western-australia.com/rock-crawling.html> - Rock crawling and rocky obstacles (including viewing of all videos linked to this site)

<http://www.iventureclub.com.au/4x4-tips/article.html?k=1ImWwpAqbd0> – 4x4 tip: Rocks and ruts

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Be prepared to ask passengers to get out of the vehicle where there are dangerous parts
- Get passengers to act as 'spotters'
- Try to stay on 'high parts' of the ground
- Re-inflate tyres when travel is completed



Slide 40

Slide No	Trainer Notes
40.	Trainer continues to discuss driving on rocky terrain: ● Be prepared to ask passengers to get out of the vehicle – if patches of the route are dangerous ● Be prepared to ask passengers to act as spotters – to assist with providing directions on where to position wheels ● Try to stay on the high parts – straddling any ruts which may be present ● Re-inflate tyres when the rocky terrain has been negotiated. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: http://www.tough toys.com.au/how-to-4x4/driving-rocky-terrain/ - Driving rocky terrain. Classroom Activity (2) – Demonstration and Practical Trainer shows students how to drive on rocky terrain and provides opportunity for them to do the same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Preparing for water crossings:

- Know when the vehicle will float
- Know how fast the water is flowing
- Know the bottom surface
- Know where the air intake
- Check out/walk the crossing
- If in doubt – wait or avoid it



Slide 41

Slide No	Trainer Notes
41.	Trainer discusses preparation for water crossings: ● Know the vehicle and how deep water the water can be before the vehicle begins to float ● Know how fast the water is flowing – to determine safety and work out speed required and angle of attack ● Know the surface to be travelled – on the bottom of the river and on exit from the crossing ● Know where the air intake is – to make sure the vehicle does not ingest water ● Check-out the crossing – walk through the crossing to determine depth, identify obstacles and work out the path to be taken ▪ Walk across the river where the left-hand wheel will go ▪ Walk back where the right-hand wheel will go ● If in doubt, avoid the crossing.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Avoid the crossing if:

- Current is too fast
- Water is too deep
- Approach angles are too steep
- Departure angle is too steep or the ground is too soft
- Vehicle is not properly prepared



Slide 42

Slide No	Trainer Notes
42.	Trainer issues standard cautions for water crossing saying got be prepared to avoid the crossing if: • The current is too fast • The water is too deep • The approach or departure angles are too steep • The approach or departure are too soft or otherwise too difficult • The vehicle is not/cannot be adequately prepared.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Making water crossings:

- Allow motor to cool
- Tie tarpaulin across front of vehicle
- Loosen fan belt
- Spray engine with water repellent

(Continued)



Slide 43

Slide No	Trainer Notes
43.	Trainer presents driving techniques for water crossings: • Let the engine cool down first – the steps below will assist with this • Tie a tarpaulin across the front of the vehicle – to help create a bow wave • Loosen the fan belt – to prevent damage to the radiator (unless vehicle is fitted with a viscous fan) • Spray the motor with water repellent (WD-40) to help preserve the electrics. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://www.iventureclub.com.au/4x4-tips/article.html?k=zFA1eN-qEzg – 4x4 Tip: Water crossing http://www.toughtoys.com.au/how-to-4x4/water-crossings/ - Water crossings

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Attach recovery straps to vehicle
- Lower vehicle windows
- Advise passengers to un-do seat belts
- Lower tyre pressures

(Continued)



Slide 44

Slide No	Trainer Notes
44.	Trainer continues to present driving techniques for water crossings: ● Attach recovery straps to the vehicle – in case there is a need to recover the vehicle from the middle of the river: make sure these straps are secured above the water line ● Lower windows – as a safety factor ● Advise passengers not to wear seat belts ● Lower pressure in tyres – if exit is sandy/soft. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=wNtJNuhlf4I – ARB Water crossing techniques http://www.outbacktravellaustralia.com.au/driving-towing-4wd-driving-skills/water-crossings - Water crossings http://www.4-wheeling-in-western-australia.com/river-crossings.html - River crossings [including videos on this page]

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Select 4WFD, Low range – 1st or 2nd gear
- Lock the diffs
- Enter at slow to moderate speed

(Continued)



Slide 45

Slide No	Trainer Notes
45.	Trainer continues to present driving techniques for water crossings: • Select 4WD, low range – first or second gear depending on type and size of engine • Lock the diffs • Enter the water at a slow to moderate speed – so as to create a bow wave but not a 'splash' which will throw water up and over the bonnet/vehicle. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://www.4wdonline.com/A/Water.html - Water crossing http://www.pps.net.au/4wdencounter/4wdtech/water.html - 4WD Encounter Australia: River crossings.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Drive slowly and evenly
- Avoid using clutch
- Dry brakes after exiting
- Re-inflate tyres



Slide 46

Slide No	Trainer Notes
46.	Trainer continues to present driving techniques for water crossings: • Drive slowly and evenly • Avoid using the clutch – as water will enter the housing and cause damage, slippage • Dry brakes after exiting the water. Classroom Activity – Demonstration and Practical Trainer shows students how to drive water crossings and provides opportunity for them to do the same.

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

Driving through mud:

- Inspect the route
- Drop tyre pressure
- Engage 4WD

(Continued)



Slide 47

Slide No	Trainer Notes
47.	Trainer discusses driving on very soft ground/mud: ● Inspecting the route – looking for: ▪ Ruts and obstacles ▪ Condition of the bottom under the surface mud/depth of the mud ▪ Distance to be covered ▪ Recovery points ● Dropping tyre pressure – to around 24 psi/165kPa ● Engaging 4WD. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://4xoverland.com/how-do-4x4s-work/4x4-driving/ - How to drive a 4x4 off road [Mud driving] https://www.youtube.com/watch?v=O7iqHO8__R8 – Mud driving the right and wrong way

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Lock diffs
- Select Low range – 1st or 2nd gear
- Enter at slow to moderate speed
- Use/do not use ruts as appropriate to surface and terrain

(Continued)



Slide 48

Slide No	Trainer Notes
48.	Trainer discusses driving on very soft ground/mud: ● Inspecting the route – looking for: ▪ Ruts and obstacles ▪ Condition of the bottom under the surface mud/depth of the mud ▪ Distance to be covered ▪ Recovery points ● Dropping tyre pressure – to around 24 psi/165kPa ● Engaging 4WD. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://4xoverland.com/how-do-4x4s-work/4x4-driving/ - How to drive a 4x4 off road [Mud driving] https://www.youtube.com/watch?v=O7iqHO8__R8 – Mud driving the right and wrong way

Slide

Drive 4WD in a variety of on-road conditions and off-road conditions

- Maintain momentum
- Spin tyres – if mud tyres are fitted
- Be prepared to reverse and try again
- Move steering from left-to-right

(Continued)



Slide 49

Slide No	Trainer Notes
49.	Trainer continues to discuss driving on very soft ground/mud: • Maintaining momentum – by applying acceleration • Spinning tyres – where mud tyres are fitted to help clean the tread • Being prepare to stop, reverse and re-try a route if progress is halted – as this may give extra traction through the more compacted bottom surface • Moving the steering wheel left-to-right – to allow side lugs of tyres to bite the mud. Classroom Activity – Demonstration and Practical Trainer shows students how to very soft ground/mud and provides opportunity for them to do the same.

Slide

Provide safety and explanatory advice to passengers

Common law imposes Duty of Care obligations on drivers to:

- Create and maintain a touring environment that does not pose a risk to people
- Take action to avoid causing foreseeable harm
- Avoid engaging in risky behaviour



Slide 50

Slide No	Trainer Notes
50.	Trainer discusses Duty of Care noting Common law imposes an obligation on all vehicle drivers to: • Create and maintain a touring environment that does not pose a risk to people (staff, members of the public, customers) • Take action to avoid causing foreseeable harm to people/customers/tour group members or their property/belongings while they are on tour • Avoid engaging in risky behaviour or actions which will pose a threat to anyone.

Slide

Provide safety and explanatory advice to passengers

Practical actions to discharge
Duty of Care obligations:

- Vehicle is safe/roadworthy before departure
- Vehicle is maintained in a safe/roadworthy condition on tour
- All road rules are obeyed
- Vehicles are driven in a safe and responsible manner

(Continued)



Slide 51

Slide No	Trainer Notes
51.	Trainer highlights in practice this means the driver needs to/has responsibility for ensuring: • The vehicle is safe/roadworthy when the tour departs • The safety/roadworthiness of the vehicle is maintained through the trip • All road rules and speed limits are obeyed • The vehicle is driven in a safe and responsible manner.

Slide

Provide safety and explanatory advice to passengers

- Obvious risks are avoided or controlled
- Passengers are advised of risks
- Passengers comply with safety requirements
- Cargo is safely stowed



Slide 52

Slide No	Trainer Notes
52.	Trainer continues discussing Duty of Care obligations imposed on tour drivers: ● Obvious risks are avoided or controlled ● Passengers are informed about potential risks/hazards ● Passengers comply with safety rules ● Cargo is stored/stowed appropriately – which may require: ▪ Re-packing/re-locating items for off-road 4-wheel drive travel to attain a lower centre of gravity for the vehicle ▪ Placing items behind physical protection barriers – or otherwise securing them to prevent injuring people or causing vehicle damage.

Slide

Provide safety and explanatory advice to passengers

Failure to meet 'Duty of Care' obligations can result in:

- Drivers being sued
- Tour Operators being sued
- The awarding of 'damages' to anyone who is injured or who has suffered loss



Slide 53

Slide No	Trainer Notes
53.	Trainer advises on consequences of failing to meet Duty of Care obligations stating if drivers of 4WDs fail to meet their Duty of Care obligations: • They may be sued by passengers/those they injure or who lose property for negligence • The Tour Operator they work for may be sued for not discharging their Due Diligence obligations by properly training their drivers • The outcomes of successful civil claims for negligence are commonly the awarding of money ('damages') to the person who has been injured or suffered loss.

Slide

Provide safety and explanatory advice to passengers

Safety information may be communicated:

- Verbally – at pre-tour and daily tour briefings
- In written form – using handouts
- By screening a safety video



Slide 54

Slide No	Trainer Notes
54.	Trainer states drivers must discuss safety with passengers and this may be done: ● Verbally – in terms of: ▪ Being part of the standard tour briefing prior to departure of the tour from the Tour Operator's office ▪ Being mentioned at the daily tour briefing (held every morning before the tour heads off for the day) on the first day there is expectation four-wheel drive will need to be engaged ● In written form – by: ▪ Providing a handout sheet ▪ Including vehicle/passenger safety as a heading in the standard tour advice/information provided to all passengers when they buy their tickets ▪ Detailing safety requirements on the business website ● By screening a video – showing: ▪ Aspects of vehicle/passenger safety ▪ Expected touring conditions ▪ How 4WDs perform in certain conditions.

Classroom Activity – Guest Speaker

Trainer arranges for a Tour Guide/Driver to attend and:

- Discuss how and when they communicate safety advice to passengers
- Identify topics which are covered by safety discussions with tour group members
- Provide examples of handouts and information supplied to tourists.

Slide

Provide safety and explanatory advice to passengers

Safety information may include advising passengers:

- To use/wear seat belts
- To stay inside the vehicle
- To keep windows wound up when travelling through bush
- To wind windows down when doing water crossing

(Continued)



Slide 55

Slide No	Trainer Notes
55.	Trainer identifies safety information may include advising passengers: ● To use/wear seat belts at all times – except when river/creek crossings are being undertaken ● To stay completely inside the vehicle at all times when the vehicle is moving – with specific reference to ensuring arms, hands and heads are kept inside the vehicle ● To keep windows wound up when travelling through bush – to prevent branches flicking people in the head as they brush against trees and bushes ● To wind windows down when undertaking river/creek crossings – to facilitate easier egress from the vehicle if there is a problem.

Slide

Provide safety and explanatory advice to passengers

- To use hand straps
- To follow driver instructions
- To not alter vehicle settings
- To secure loose personal items

(Continued)



Slide 56

Slide No	Trainer Notes
56.	Trainer continues to identify safety information may include advising passengers: • To use hand straps/holds when the vehicle is traversing uneven ground • To follow the driver's instructions/directions at all times – fully and promptly • To refrain from altering any vehicle settings – or tampering with/touching any of the controls of the vehicle • To secure loose personal items when the vehicle is in motion – advising how and where this may be done.

Slide

Provide safety and explanatory advice to passengers

- To not throw rubbish from vehicle
- To refrain from poor/distracting behaviour
- To wait until vehicle stops before getting on/off
- About on-board emergency items and recovery gear
- About EMPs for the tour/trip



Slide 57

Slide No	Trainer Notes
57.	Trainer continues to identify safety information may include advising passengers: ● To refrain from throwing items/litter from the vehicle ● To refrain from behaviour that would be distracting to the driver of the vehicle ● To only get in and out of the vehicle when it has come to a full stop ● About on-board emergency items, first aid kits, recovery equipment, communication systems and emergency exits ● About Emergency Management Plans which have been developed – to guide action in the event of an emergency or accident. Classroom Activity – Handouts Trainer distributes and discusses sample EMPs.

Slide

Ensure minimal environmental impact when driving a 4-wheel drive vehicle

4WDs may cause environmental damage by:

- Disturbing local communities
- Damaging/destroying flora
- Causing noise pollution

(Continued)



Slide 58

Slide No	Trainer Notes
58.	Trainer explains it is possible 4WDs may impact the environment in the following ways: • Disturbing local communities – and other visitors in the area • Damaging/destroying flora – which can also lead to erosion of the land • Causing noise pollution.

Slide

Ensure minimal environmental impact when driving a 4-wheel drive vehicle

- Contaminating water
- Harming soil structure
- Scaring and/or injuring wildlife



Slide 59

Slide No	Trainer Notes
59.	Trainer continues to identify possible environmental damage caused by 4WDs: • Contaminating fresh water sources • Harming soil structure – by compacting, dislodging it and/or causing pot-holes • Scaring, injuring or killing local wildlife.

Slide

Ensure minimal environmental impact when driving a 4-wheel drive vehicle

Techniques to minimise environmental damage:

- Adhering to designated tracks
- Assessing quality of tracks before travelling on them
- Minimising the effects of noise
- Ensuring required permits have been obtained

(Continued)



Slide 60

Slide No	Trainer Notes
60.	Trainer presents techniques to minimise environmental damage: ● Adhering to designated tracks and routes in sensitive areas, parks or other prescribed areas – as opposed to creating new tracks and driving off-road ● Assessing quality of tracks or intended routes before travelling on them – and not travelling on tracks which are: ▪ Closed ▪ Degraded or boggy ● Minimising the effects of noise – by: ▪ Ensuring exhaust and muffler system is effective ▪ Avoiding revving/racing engines ▪ Staying clear of other users ● Ensuring all required permits (including payment of fees), authorities and advice have been obtained – and their requirements have been understood and complied with.

Slide

Ensure minimal environmental impact when driving a 4-wheel drive vehicle

- Abiding by local prohibitions
- Being vigilant at all times
- Ensuring passenger behaviour is proper
- Being patient

(Continued)



Slide 61

Slide No	Trainer Notes
61.	Trainer continues to present techniques to minimise environmental damage: ● Abiding by local prohibitions – regarding, for example: ▪ Entry by vehicles into nominated areas ▪ Use of tracks under certain weather conditions ● Being vigilant when preparing, planning and driving – so as to avoid causing damage to: ▪ Property and structures ▪ Natural habitats of animals ▪ Natural assets ● Ensuring proper behaviour of passengers – in relation to applicable Codes of conduct which may require: ▪ They do not litter the route and areas travelled ▪ They do not remove flora and/or fauna from the area ▪ They do not use firearms

- | | |
|--|---|
| | <ul style="list-style-type: none">• Being patient – and:<ul style="list-style-type: none">▪ Not speeding▪ Using passing bays, where provided, rather than driving off-road and creating new tracks in order to get passed other vehicles▪ Taking the time to close gates which had to be opened – standard protocol is to leave all gates as they were found. |
|--|---|

Slide

Ensure minimal environmental impact when driving a 4-wheel drive vehicle

- Removing obstructions from tracks rather than driving around them and creating fresh tracks
- Not using trees as recovery points if alternatives exist
- Giving way to foot traffic and animals

(Continued)



Slide 62

Slide No	Trainer Notes
62.	Trainer continues to present techniques to minimise environmental damage: ● Removing obstructions on tracks – as opposed to driving round them and creating new tracks ● Not using trees as winch points if a recovery is required – or using tree protectors if no other suitable point is available ● Giving way to foot traffic on tracks – and allowing animals to have right of way.

Slide

Ensure minimal environmental impact when driving a 4-wheel drive vehicle

- Driving on sand below high water marks
- Adhering to weight limits
- Taking care not to start fires when driving on grass
- Not travelling at night



Slide 63

Slide No	Trainer Notes
63.	Trainer continues to present techniques to minimise environmental damage: ● Driving on beaches only on the firm sand below high water marks – so high tide will remove vehicle marks ▪ Note: while driving below the high tide mark will reduce 4WD impact many people recommend this is not done in case the vehicle becomes stuck/breaks down in which case it may be swamped with water when the tide comes in ● Adhering to weight limits for vehicles and trailers – to avoid extra depression of ruts/soil ● Taking care when driving through dry grass – that the vehicle's exhaust system does not start fires ● Not travelling at night – when there is greater danger of hitting wildlife.

Slide

Comply with road safety and road use rules and regulations

Compliance requirements imposed on drivers include:

- Be properly licensed
- Carry their licence
- Ensure vehicle is registered
- Display/carry proof of registration

(Continued)



Slide 64

Slide No	Trainer Notes
64.	Trainer explains drivers of four-wheel drive vehicles are required to: ● Be properly licensed – according to: ▪ The size of the vehicle ▪ Type/class of vehicle ▪ Whether it is manual or automatic ● Carry their licence – and present it on demand to authorities ● Ensure the vehicle is correctly registered – in accordance with: ▪ The type/class of vehicle being used ▪ The commercial nature of the vehicle ● Display proof of registration – to demonstrate current and relevant registration.

Slide

Comply with road safety and road use rules and regulations

- Ensure vehicle compliance with roadworthiness
- Obtain and display relevant permits
- Comply with road rules

(Continued)



Slide 65

Slide No	Trainer Notes
65.	Trainer continues to identify obligations imposed on drivers: ● Ensure the vehicle complies with roadworthy requirements – as these apply to: ▪ Vehicle safety ▪ Passenger safety ● Obtain, display and comply with applicable special permits as/if required by local laws – for use of designated: ▪ Parks ▪ Tracks ▪ Local roads ▪ Beaches ▪ Other routes/locations ● Comply with all applicable road rules – as these apply to: ▪ Adhering to speed limits ▪ Obeying traffic signs ▪ Giving right of way to pedestrians and other vehicles – and passing and overtaking ▪ Using indicators, lights and warning and safety devices ▪ Demonstrating consideration and tolerance for other road users.

Slide

Comply with road safety and road use rules and regulations

- Comply with fatigue management protocols
- Comply with environment protection legislation
- Comply with drink-driving and drug-related requirements



Slide 66

Slide No	Trainer Notes
66.	Trainer continues to identify obligations imposed on drivers: ● Comply with applicable fatigue management regulations – so the driver has sufficient: ▪ Rest/sleep ▪ Breaks from driving ● Comply with applicable environmental protection legislation – as this applies to: ▪ Smoke emissions ▪ Noise pollution ● Comply with relevant drink-driving legislation – and does not drive contrary blood alcohol limits as these apply to: ▪ Age of the driver ▪ Licence status ▪ Type/class of vehicle being driven ● Comply with legislation prohibiting driving while under the influence of drugs.

Classroom Activity – Guest Speaker

Trainer arranges for official from local Transport Agency to attend and:

- Discuss driver licensing and vehicle registration requirements for tour 4WDs
- Explain ways in which Tour Operators and 4WD drivers can optimise compliance with obligations
- Describe consequences for non-compliance
- Provide assistance to enable/encourage compliance.

Slide

Element 2 – Recover a 4WD vehicle

Performance Criteria for this Element are:

- Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations
- Optimise safety for passengers and protection of property
- Inspect vehicle after recovery and perform necessary maintenance and repairs before proceeding



Slide 67

Slide No	Trainer Notes
67.	Trainer identifies for trainees the Performance Criteria for this Element, as listed on the slide. Class Activity – General Discussion Trainer leads a general class discussion by asking questions such as: • What does 'recovery of a vehicle' mean? • What situations might give rise to the need for a recovery? • How might recoveries be undertaken? • What experience do you have with recovering vehicles?

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Golden Keys to successful vehicle recovery:

- Practice
- Experience



Slide 68

Slide No	Trainer Notes
68.	Trainer identifies Golden Keys to successful recovery of four-wheel drive vehicles are the two factors identified earlier: • Practice • Experience. It is critical drivers of 4WD tour vehicles gain competency in vehicle recovery in all expected on- and off-road conditions <i>before</i> they do so in a real-life touring environment where paying passengers/tour groups are involved.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Vehicles may need to be recovered from:

- Sand
- Mud
- Water
- Rocky terrain



Slide 69

Slide No	Trainer Notes
69.	Trainer advises there can be a need to recover vehicles from situations where they have become bogged, beached or otherwise trapped: ● In sand ● In mud/soft ground ● In water ● On rocky surfaces.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Vehicle driver must:

- Know the recovery gear provided with the vehicle
- Know where in/on the vehicle the recovery gear is located
- Know the mounting points on the vehicle for recovery gear
- Prepare/attach recovery gear in advance
- Check recovery gear for safety and readiness



Slide 70

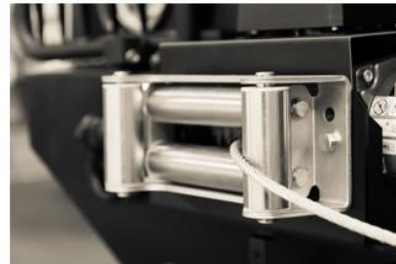
Slide No	Trainer Notes
70.	Trainer discusses recovery gear knowledge stressing vehicle driver must: • Be aware of the gear available on/in the vehicle for recovery situations • Know where the recovery gear is located • Know the recovery points/hooks and mounting features/points of the vehicle – which are connected directly to the vehicle chassis to give optimum strength • Prepare/attach any gear which is deemed necessary/possibly required for an upcoming situation • Check the gear for safety and readiness for use – especially: ▪ Checking for frayed cables and straps ▪ Verifying winches which have been immersed in water on previous tours have been checked and serviced ▪ Making sure items which may be required are readily accessible.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Recovery gear must be correctly 'rated' so it will withstand weights to be applied during recoveries – with reference to:

- Snapping breaking of items especially shackles and cables, ropes and straps
- Jacks
- Winches



Slide 71

Slide No	Trainer Notes
71.	Trainer discusses 'rating' of recovery gear stating: ● It is important to use recovery gear that is rated correctly for the vehicle and the recovery situation ● If incorrect/insufficiently rated gear (cables, straps, mounting points, winches, jacks) is used there is danger posed through: ▪ The snapping/breaking of items ▪ Items being ripped off vehicles ▪ Jacks collapsing under weight ▪ Winches failing under load ● Rating refers to the weight (in tons or tonnes) of the vehicle being recovered known as Minimum Breaking Strength (MBS) ● Manufacturers may also rate their products in generic classifications such as: ▪ Standard ▪ Heavy Duty ▪ Extra heavy duty.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Recovery gear might include:

- Long-handled shovel
- Tyre deflators
- Shackles
- Drag chains

(Continued)



Slide 72

Slide No	Trainer Notes
72.	Trainer identifies recovery gear may include: • A shovel – traditionally a basic long-handled shovel • Tyre deflators – to lower tyre pressures before attempting to drive out of a bogged/trapped situation • Rated shackles ('D' shackles or 'Bow' shackles) appropriately rated to the weight to be handled – used to attach recovery cables/straps to vehicle mounting points and to attach snatch straps to tree protectors. Never use these to connect two snatch straps together • Drag chains – used to tow another vehicle or to remove (for example, fallen trees) from the track. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=CDqu3La1nls – 4WD Recovery series Part 1: The essential gear Classroom Activity (2) – Samples Trainer shows students examples and explains use of recovery items identified on the slide.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Snatch straps
- Winches
- Inflatable bags
- Jacks
- Recovery tracks/sand ladders

(Continued)



Slide 73

Slide No	Trainer Notes
73.	Trainer identifies recovery gear may include: • Snatch straps – which are used in combination with another vehicle to extricate ('snatch') a trapped vehicle free from sand or mud • Winches and cables – including electric winches, PTO winches, hydraulic winches, capstan ('wheel') winches, hand-operated winches and snatch blocks • Inflatable bags – inflated by pumps or the exhaust • Jacks – including standard scissor and bottle jacks as well as high-lift jacks (also known as 'kangaroo jacks') • Recovery tracks and/or sand ladders – for sand and bog recoveries as well as for bridging gaps. Classroom Activity – Samples Trainer shows students examples and explains use of recovery items identified on the slide.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Accessories:

- Winch gloves
- Winch extensions
- Extension straps
- Snatch strap joiners
- Equaliser straps
- Cable blankets/dampers
- Ground anchors



Slide 74

Slide No	Trainer Notes
74.	Trainer identifies recovery gear may include: ● Accessories – such as: ▪ Winch gloves – to protect hands when handling winch cables ▪ Winch extensions – to give extra length to winch cables ▪ Extension straps – to give extra length to snatch straps ▪ Snatch strap joiners – to assist in the joining of snatch straps so they can be easily disconnected ▪ Tree protectors – to wrap around tress to protect them when winching out ▪ Equaliser straps – effectively allows snatch straps to be connected to two recovery points on the vehicle being recovered instead of just one thereby spreading the load ▪ Recovery blankets/dampers (also known as a cable dampener) – placed over cables and snatch straps when winching or snatching to absorb energy and protect by-standers in the event a winch cable or strap breaks when under load ▪ Ground anchors – used to help recover a vehicle where there are no natural recovery points/trees to attach a winch cable to. Classroom Activity – Samples Trainer shows students examples and explains use of recovery items identified on the slide.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Long-handled shovels may be used for:

- Digging away sand or dirt
- Getting rid of debris from tracks/routes
- Removing unwanted soil or material from approach and departure angles



Slide 75

Slide No	Trainer Notes
75.	Trainer discusses recovery techniques: Using long-handled shovel The long-handled shovel is a basic recovery tool often used as a first-response when vehicles are stuck, or to even out tracks to prevent vehicles <i>becoming</i> trapped. They are used for: ● Digging away sand or dirt – from under the wheels of vehicles ● Getting rid of debris from tracks/routes ● Removing unwanted soil or material from approach and departure angles of river crossings and steep inclines. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://www.iventureclub.com.au/4x4-tips/article.html?k=cSwbJ7Nu3UQ 4x4: Solo recovery.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Using a snatch strap:

- Attach strap to vehicle mounting point/s
- Unroll strap/s between vehicles
- Add extra strap – if more length is required
- Attach strap to retrieving vehicle



(Continued)

Slide 76

Slide No	Trainer Notes
76.	Trainer discusses snatch strap recovery: ● Attaching strap to vehicle mounting/recovery points only on both vehicles – and not to tow ball, bull bar or bumper bar ● Unrolling the strap between the two vehicles making sure the strap is straight, not twisted and clear between the vehicles – and is not kinked or blocked by obstacles ● Adding another snatch strap – where extra length is required between the recovery vehicle and the trapped/bogged vehicle ● Attaching strap to vehicles: ▪ Using a rated shackle – as opposed to simply looping over an attachment point ▪ Securing to central mounting/recovery points on both vehicles. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=-lre31FrEAI – Snatch strap recovery technique https://www.youtube.com/watch?v=bRRhfz4tdRg – Ask an expert 031: Attaching a snatch strap

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Use shovel to remove build-up from in front of wheels
- Position recovery vehicle in front of trapped vehicle
- Place damping blanket over strap
- Ensure comms between drivers of two vehicles
- Start trapped vehicle – 4WD, Low range, first gear



(Continued)

Slide 77

Slide No	Trainer Notes
77.	Trainer continues to describe snatch strap recovery: ● Using long-handled shovel to remove build up in front of the wheels – in the direction the recovery will occur ● Position the recovery vehicle so there is about three metres of slack in the snatch strap – to allow for a run-up to 'snatch' the trapped vehicle from its position ● Placing recovery damper/blanket across snatch strap in the middle third part of the strap – for safety reasons ● Ensuring effective communication between the recovery vehicle and the trapped vehicle – so the two drivers can coordinate their actions ● Starting the trapped/vehicle and engaging 4WD, Low range, first gear. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=oT6R6yTg6R0 – Mad Matt's Snatch Strap Recovery Tips https://www.youtube.com/watch?v=i7GSL3o1pUU – 4WDRecovery series Part 2: Snatch strap recovery technique

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Confirm 'good to go'
- Drive recovery vehicle forward at medium/moderate speed – 4WD, Low range, 2nd gear
- Accelerate trapped vehicle to assist recovery
- Repeat if necessary



Slide 78

Slide No	Trainer Notes
78.	Trainer continues to describe snatch strap recovery: ● Confirming between drivers their readiness to start the procedure ● Driving the recovery vehicle forward at a medium/moderate rate in 4WD, Low range, second gear – as opposed to racing forward quickly at high speed ● Accelerating the trapped vehicle so it can assist the recovery vehicle when the snatch strap takes effect ● Realising the vehicle does not have to be recovered/freed on the first attempt – it is possible/may be necessary to repeat and repeat to free the trapped vehicle. Classroom Activity – Demonstration and Practical Trainer shows students how to recover a vehicle using a snatch strap and provides an opportunity for students to do the same.

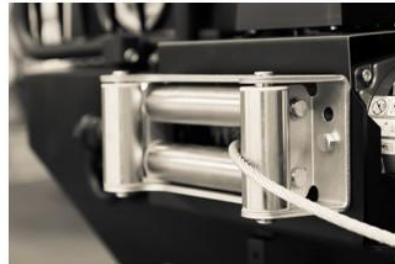
Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Using electric winch:

- Always wear gloves when using winch
- Follow winch manufacturer's instructions
- Unwind cable off the winch
- Place tree protector around tree to be used as recovery point

(Continued)



Slide 79

Slide No	Trainer Notes
79.	Trainer discusses recovery techniques: Using electric winch • Always wear gloves • Always follow manufacturer's instructions – especially as these apply to: ▪ Safety issues ▪ Operation of the winch ▪ Use of attachments • Unwind the cable off the winch – giving sufficient length to attach to the recovery point: add extra lengths of cable if necessary • Place tree protector around the tree – and attach the winch cable with a rated shackle. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=AKLtDFz5WgU – 4WDRecovery series Part 3: Basic winching techniques https://www.youtube.com/watch?v=L2l_-3lpHII – 4WDRecovery series Part 4: Advanced winching techniques

	https://www.youtube.com/watch?v=StZHIHD09XM – 4x4 The basic guide to winching techniques https://www.youtube.com/watch?v=jatVOGInI64 – 4WD winching techniques with Pat Callinan – Ray's Outdoor
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Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Put winch/cable dampener over cable
- Take up the slack on the cable
- Verify passengers are out of danger ('Danger Zone')
- Start recovery by engaging winch

(Continued)



Slide 80

Slide No	Trainer Notes
80.	Trainer continues to describe electric winch recovery: ● Drape a winch dampener over the cable – as a safety precaution ● Take up the slack on the cable between the tree and the trapped vehicle – guide the cable as it winds back on to the winch to ensure it rolls on evenly and smoothly ● Verify passengers are out of danger ● Start the recovery by engaging the winch – opinions vary and there are two basic options: ▪ Allow the winch to do all the work and recover the vehicle on its own ▪ Put the vehicle in 4WD Low range first or second gear and gently accelerate to assist the action of the winch. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=StZHIHD09XM – The basic guide to winching techniques [for electric winches] https://www.youtube.com/watch?v=F4mXgnll3e4 – How to winch a 4x4 https://www.youtube.com/watch?v=_4tl3HtgdcY – Reverse winch https://www.youtube.com/watch?v=Td92wGxvujU – Snatch block in action

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Monitor take up of cable on spool of winch
- Operate winch to preserve battery of vehicle
- Use pulley/snatch blocks to assist if necessary



Slide 81

Slide No	Trainer Notes
81.	Trainer continues to describe electric winch recovery: ● Have person (tour crew member) monitor the take up of the cable on the winch spool/drum – to ensure an even wind on ● Be aware of the heavy draw on electric power of the winch – so: ▪ Leave engine running throughout the recovery operation ▪ Minimise its use ▪ Be prepared to use it in small bursts, stopping in between to run the engine and re-charge the battery ● Use a pulley block at the anchor point ('double line pull') to assist where necessary – which will halve the load on the winch: put a dampener over both cables. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=JY4cKc4FKSM – Going 4WD: The Bush Winch/Wheel Winch http://www.outbacktravellaustralia.com.au/driving-towing-recovery-techniques/safe-winchng - Safe winching https://www.youtube.com/watch?v=4TtAgDp6VKA – The intelligent winch: 4x4.

	Classroom Activity (2) – Demonstration and Practical
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	Trainer shows students how to recover a vehicle using an electric winch and provides an opportunity for students to do the same.
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Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Alternate recovery winches include:

- PTO winch
- Hydraulic winch
- Capstan winch



Slide 82

Slide No	Trainer Notes
82.	Trainer identifies alternate recovery winches: The PTO winch A PTO winch is driven by engine of 4WD and operates winch drum/spool through gearbox via a power take-off. PTO winches are also mounted at the front of the vehicle. They are a good option where long, heavy winching work is expected but have one major short-coming in that they cannot be used if the engine of the vehicle is dead/will not start. The same protocols described for the use of the electric winch apply to the use of the PTO winch. Hydraulic winch Hydraulic winches are powered by a hydraulic motor/pump which is, in turn, powered by the engine of the vehicle. This style of winch is not a common option but seen by many as one of the best options in that it: • Allows the same control over the winch as for an electric winch • Does not pose the same problem with flattening the battery so it can be used for extensive lengths of time

- The down-side is the engine of the 4WD needs to be operational meaning it will not work if the engine cannot be started.

Capstan winch

A capstan winch features a drum/spool plus a capstan around which cable/winch rope is fed when recovering the vehicle.

The use of a capstan means vehicle recovery points can be 'at an angle' to the vehicle without this causing a problem.

They are generally powered off the engine of the vehicle and not via an electric motor.

Classroom Activity – Internet research

Trainer supervises internet research on relevant websites such as:

<https://www.youtube.com/watch?v=xDAzHXkk3LE> – PTO winches

<https://www.youtube.com/watch?v=GOMZQldEctg> – Red Warrior hydraulic winch with PTO power

<http://www.4wdonline.com/Equip/winches.html> - 4WD Online: Winches

<http://www.4wdonline.com/Equip/winch-TJM.html> - TJM hydraulic winch

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Hand-operated winches are available but not commonly used because:

- They are bulky to carry
- They take time to use
- They have limited cable length



Slide 83

Slide No	Trainer Notes
83.	Trainer discusses recovery techniques: Hand-operated winch Some vehicles may carry a hand-operated winch but they are uncommon as: • They are bulky to carry • Take more time to use when recovering a vehicle than other options • Are limited in the length of cable that is available for use in a recovery.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Using a hand-operated winch:

- Follow manufacturer's instructions
- Wear gloves
- Attach cable hook to vehicle mounting point on trapped vehicle
- Run out cable to recovery point

(Continued)



Slide 84

Slide No	Trainer Notes
84.	Trainer describes technique for operating hand winch for vehicle recovery: ● Follow manufacturer's instructions ● Wear gloves ● Attach cable hook to recovery mounting on vehicle ● Run out the cable to the tree which is to be used as the recovery point. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=5UzX3F4Oyqw – 4WD Action: 114 Terrain Tamer Hand Winch

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Attach tree protector
- Thread cable through winch
- Attach winch to tree protector with shackle
- Clamp winch onto cable

(Continued)



Slide 85

Slide No	Trainer Notes
85.	Trainer continues to describe technique for operating hand winch for vehicle recovery: • Attach tree protector to recovery point with rated shackle • Thread cable through the winch – setting levers in the correct position to allow the cable to be fed through the winch • Attach the winch to the tree protector using the shackle • Clamp the winch onto the cable – by moving the appropriate lever on the winch. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=k39wFI4MQQg – ask an expert: hand winching

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Drape dampener/blanket over cable
- Insert handle into winch
- Pump/move handle to start the winching process



Slide 86

Slide No	Trainer Notes
86.	Trainer continues to describe technique for operating hand winch for vehicle recovery: ● Drape cable damper over the winch cable ● Insert handle into winch ● Pump/move the handle to commence the winching process. Note: snatch blocks/pulleys can be used in combination with hand winches in the same way they can be used with other winches. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=MvU2g4dyKhl – Exacta hand winch Classroom Activity (2) – Demonstration and Practical Trainer shows students how to recover a vehicle using a hand-operated winch and provides an opportunity for students to do the same.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Using hi-lift jack:

- Spray with WD-40
- Follow manufacturer's instructions
- Fit jack into base plate
- Provide solid surface for lift – to best extent possible

(Continued)



Slide 87

Slide No	Trainer Notes
87.	Trainer describes technique for using hi-lift jack for vehicle recovery: • Clean and spray with WD-40 – as these jacks are usually carried on the outside of the vehicle and become dirty • Follow manufacturer's instructions • Fit jack into base plate – if base is separate • Provide solid surface on which to locate the base of the jack – as best as possible given the conditions. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=DgjqhU5PgkE – ARB hi-lift jack demo https://www.youtube.com/watch?v=1a_5lpDzPPI – How to use a hi-lift jack safely

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Flip out lift plate/bar
- Position jack and fit/fix lift plate/bar to vehicle
- Move selector to 'Lift'
- Insert jack handle – lift slowly keeping head out of the way

(Continued)



Slide 88

Slide No	Trainer Notes
88.	Trainer continues to describe technique for using hi-lift jack for vehicle recovery: ● Flip out the lift plate/bar on the jack – which will be used to take the weight when the vehicle is lifted ● Locate jack as required near lifting point, which may be: ▪ Solid part of the vehicle such as a steel bull bar ▪ A hole/lug into which the lift bar is inserted ▪ Adjacent to the bogged wheel – where a lift strap ('lift mate') can be attached to the wheel, instead of to a vehicle lift point ● Move selector/lever to 'lift' ● Insert jack handle and pump to lift vehicle – taking care: ▪ To lift slowly ▪ To use both hands to optimise control over the handle ▪ To not allow head to be positioned above the handle – which can kick back and cause injury to the operator ▪ To keep the jack stable – and not allow it to topple.

Classroom Activity – Internet research

Trainer supervises internet research on relevant websites such as:

<https://www.youtube.com/watch?v=hUL-l9Yi3Ws> – How to use the high-lift jack

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Apply pressure at tip of handle for maximum leverage
- Listen for 'click' as lift climbs
- Lift vehicle clear of obstruction
- Insert rocks, timber, tracks or sand ladders
- Lower jack – selecting 'Down' and operating slowly and using both hands



Slide 89

Slide No	Trainer Notes
89.	Trainer continues to describe technique for using hi-lift jack for vehicle recovery: ● Apply pressure to the end of the handle – for maximum leverage ● Listen for the 'click' of the mechanism to indicate end of one stroke and need to start next pump ● Lift vehicle until it is clear of obstruction/out of the trapped situation ● Insert rocks, wood or proprietary products (portable tracks and/or sand ladders) under wheel and in direction of travel ● Lower jack – by: ▪ Shifting lever to reverse position ▪ Pumping the handle slowly ▪ Keeping head away from handle ▪ Taking greater care as the handle/jack is not fully weighted and can provide a greater danger of causing injury should it become out-of-control during the lowering process.

Classroom Activity (1) – Internet research

Trainer supervises internet research on relevant websites such as:

<https://www.youtube.com/watch?v=9FRMK7ZstaY> – High lift jack recovery: off road recovery

Classroom Activity (2) – Demonstration and Practical

Trainer shows students how to recover a vehicle using a hi-lift jack and provides an opportunity for students to do the same.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Note:

- It is also possible to use the hi-lift jack as a winch to recovery a vehicle if no other 'proper' or more suitable winch is available



Slide 90

Slide No	Trainer Notes
90.	Trainer advises it is also possible to use the high-lift jack as a hand winch for some recoveries. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=OVY0QQiN93o – Using high-lift jack as winch to get Discovery out https://www.youtube.com/watch?v=bOcQHqLx90g – Hi-lift as a winch https://www.youtube.com/watch?v=pAGp8lizSQU – Basic self recovery using a hilit. Classroom Activity (2) – Demonstration and Practical Trainer shows students how to use hi-lift jack as a recovery winch to recover a vehicle and provides an opportunity for students to do the same.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Inflatable bags may be used to recover trapped vehicles – by:

- Using exhaust gases of the vehicle
- Using an air compressor.
- Once the vehicle is raised, other items will be required to finish the recovery



Slide 91

Slide No	Trainer Notes
91.	Trainer explains under-vehicle inflatable bags are available to help raise bogged/trapped vehicles, or to use an alternative to a scissor or bottle jack when changing a wheel and these bags may be inflated: ● Using the exhaust gases of the vehicle/engine ● Using an air compressor. They need to be used in combination with rocks, wood or proprietary products (portable tracks and/or sand ladders) under wheel/s and in direction of travel for them to be really useful: simply lifting a trapped vehicle is not of much use – it needs to be able to move forward and the rocks, wood, tracks or ladders will allow this to happen. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=aILTIKpu3Zs – Exhaust jack

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Using inflatable bag:

- Unpack bag from carrier
- Locate under the vehicle in appropriate position
- Check to ensure bag will not be damaged by the vehicle/the lift
- Attach inflation hose to bag

(Continued)



Slide 92

Slide No	Trainer Notes
92.	Trainer gives instruction on how to use inflatable bag to lift trapped vehicle: ● Unpack the bag from its carrier ● Locate in appropriate position under the vehicle ● Check to ensure: ▪ No jagged edges/protrusions which can harm or hoe the bag ▪ Bag is away from hot exhausts and engine parts ● Attach hose to bag. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=CqBQ8q2mxuY – X-Jack and X-Trax

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Start vehicle engine
- Attach cone of inflation hose to end of exhaust pipe
- Watch as bag inflates

(Continued)



Slide 93

Slide No	Trainer Notes
93.	Trainer continues to describe use of inflatable bag to lift vehicle: • Start vehicle engine in neutral – if using exhaust to lift the vehicle • Take other end of hose to exhaust pipe and jam/hold the cone of this end in position over end of the exhaust pipe • Watch as bag inflates. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=Hc7Tni99dlQ – LROTV High lift jack versus Air jack

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Remove hose from exhaust when bag inflates sufficiently
- Insert rocks, wood, tracks or ladder under wheel
- Loosen hose connected to bag to lower vehicle



Slide 94

Slide No	Trainer Notes
94.	Trainer continues to describe use of inflatable bag to lift vehicle: ● Remove hose from exhaust pipe when bag is inflated ● Insert rocks, wood or proprietary products (portable tracks and/or sand ladders) under wheel and in direction of travel ● Loosen hose connected to bag – to deflate the bag slowly and lower the vehicle. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=gmzMx9_TkaU – How to use an exhaust air-jack for recovery. Classroom Activity (2) – Demonstration and Practical Trainer shows students how to lift a vehicle using an inflatable bag and provides an opportunity for students to do the same.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Sand ladders and other proprietary vehicle tracks can be carried and can be:

- Placed under wheel to give better purchase when trying to recover a vehicle
- Used to build a 'bridge' (or ramp) to help cover otherwise insurmountable rises or gaps

(Continued)



Slide 95

Slide No	Trainer Notes
95.	Trainer introduces use of sand ladders and other proprietary light-weight tracks for four-wheel driving and recovery observing they can be: ● Placed under the wheels of stranded vehicles to give the wheels better purchase when trying to un-stick the vehicle by either towing/winching it out or driving it out ● Used to provide a bridge to allow a vehicle to travel across an otherwise insurmountable gap. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=NLe71YjrUkl – Muputrax: multi-purpose recovery tracks/sand ladders https://www.youtube.com/watch?v=kMFa_pylslo – LROTV: Sand ladders demonstration

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

Using sand ladders/tracks:

- Shovel away build-up from in front of tyres
- Raise the vehicle/jack up the vehicle
- Position ladders/tracks in front of tyres

(Continued)



Slide 96

Slide No	Trainer Notes
96.	Trainer describes use of sand ladders/tracks: ● Shovel away build up from in front of the tyre/s – if possible ● Raise the vehicle – if possible ● Position the ladders/tracks under and in front of the wheels. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=95m6VHEWvFk – Overland journal: sand ladder test https://www.youtube.com/watch?v=hm_u0wb3kD0 – Terra Trax sand ladders demo.

Slide

Use a variety of techniques to recover 4-wheel drive vehicles

- Lower vehicle onto ladder/tracks
- Engage 4WD, Low range, 1st gear
- Drive slowly forward
- Repeat placement of ladders/tracks as many number of times as needed



Slide 97

Slide No	Trainer Notes
97.	Trainer continues to describe use of sand ladders/tracks: • Lower the vehicle onto the ladders/tracks • Make sure vehicle is in 4WD Low range first gear • Drive forward • Repeat as required until vehicle is free. Classroom Activity – Demonstration and Practical Trainer shows students how to use sand ladders/tracks to recover a trapped vehicle and provides an opportunity for students to do the same.

Slide

Optimise safety for passengers and protection of property

- Vehicle recoveries often pose risks to passengers and property
- Dangers may relate to:
 - Risk of vehicle being flooded if trapped in tidal river or beach
 - Risk of vehicle slipping, rolling, toppling or moving unexpectedly

(Continued)



Slide 98

Slide No	Trainer Notes
98.	Trainer highlights recovering a trapped vehicle can present various risks/dangers depending on where and how the vehicle is stuck with possible dangers including: ● Risk of the vehicle becoming flooded in a tidal river or beach situation as the water rises ● Danger of the vehicle slipping and rolling forward, backwards or sideways.

Slide

Optimise safety for passengers and protection of property

- Chance of vehicle trapped in water being pushed over/swept away
- Winch cable/recovery rope or strap breaking and springing back to injure people or damage property
- Passengers leaving a stranded vehicle



Slide 99

Slide No	Trainer Notes
99.	Trainer continues to identify possible dangers when recovering vehicles: ● Chance of vehicle trapped in water being pushed over by the force of the water (current/flow) against its side ● Possibility of winch cable, snatch strap or tow rope breaking and springing back causing injury to people in vehicle or by-standers ● Passengers injuring themselves as they leave/get out of a stranded vehicle – and slip/slide on muddy, wet or otherwise slippery surfaces.

Slide

Optimise safety for passengers and protection of property

Recovery 'Danger Zone':

- Area in which passengers/people may be injured when vehicle recovery is being undertaken if something goes wrong
- SOP is:
 - Remove everyone from the Danger Zone
 - Protect them behind a solid barrier



Slide 100

Slide No	Trainer Notes
100.	Trainer advises it is useful to consider the concept of 'Danger Zone' when recovering a trapped vehicle. ● The Danger Zone is the area in which passengers/people could be injured if something goes wrong with the recovery ● Standard practice is to consider the factors which impact on defining the Danger Zone for each recovery situation and then: ▪ Move everyone out of that zone ▪ Position them behind walls, rocks or other suitable protection.

Slide

Optimise safety for passengers and protection of property

Factors influencing determination of Danger Zone:

- Type of recovery gear being used
- Direction of recovery to be attempted
- Potential for vehicle to roll, slip, topple or fall
- Effect of suction on cable/winch capacity
- Level of difficulty of the recovery



Slide 101

Slide No	Trainer Notes
101.	Trainer identifies factors to be considered when determining the Danger Zone for a recovery can include: • Type of recovery equipment used – which indicates possibility of (cable or rope or other) failure/s • Direction of recovery to be attempted for the trapped vehicle • Potential for vehicle to roll, slip, topple or fall – which may be related to: ▪ Fall/lay of the land ▪ Surface conditions ▪ Physical position of the vehicle • Effect of suction on cable/winch capacity • Level of difficulty of the recovery.

Slide

Optimise safety for passengers and protection of property

'People safety is paramount' during all recoveries – this means:

- Doing nothing to jeopardise safety of humans
- People take priority over property

(Continued)



Slide 102

Slide No	Trainer Notes
102.	Trainer stresses the Golden Rule when conducting a recovery or driving a four-wheel drive is 'people safety is paramount' explaining this means: • Doing nothing to jeopardise the safety of humans • Safety of people comes before protecting their property.

Slide

Optimise safety for passengers and protection of property

- Passengers safety takes precedence over animals
- Passengers take priority over the tour vehicle
- The itinerary must not determine safety decisions and actions
- Financial considerations must not dictate action taken to ensure safety



Slide 103

Slide No	Trainer Notes
103.	Trainer continues to present practical aspects of passenger safety when doing a vehicle recovery: • Safety of people comes before the safety of animals • Passengers must be protected before the safety of the tour vehicle • Not allowing time pressures of a tour itinerary to impact on decisions and actions required to safely address a recovery situation • Not allowing financial considerations to dictate an unsafe course of action is chosen over a known, more expensive but safer recovery option.

Slide

Optimise safety for passengers and protection of property

Preparation for recovery may include:

- Check and prepare recovery gear
- Check/inspect the recovery route
- Check/re-locate/secure items in the vehicle
- Advise passengers of identified threats
- Give necessary safety advice



Slide 104

Slide No	Trainer Notes
104.	Trainer identifies preparation for situations where the vehicle could become trapped may include: ● Check and prepare recovery gear which may be required – to ensure it is: ▪ Quickly available ▪ Fit for use should it be needed ● Check out/walk or otherwise inspect the route ● Re-locate items in the vehicle – in order to: ▪ Alter the balance/weight distribution ▪ Secure items ▪ Give ready access to certain gear ● Advise passengers of identified threats ● Give necessary safety advice – such as: ▪ Wind windows down ▪ Keep doors unlocked ▪ Keep seat belts on or off – depending on the situation.

Slide

Optimise safety for passengers and protection of property

If tour vehicle becomes trapped/stuck, driver and/or tour crew must:

- Take control
- Show leadership and display confidence
- Avoid panic

(Continued)



Slide 105

Slide No	Trainer Notes
105.	Trainer identifies if the tour vehicle becomes trapped the driver/tour crew must: • Take control of the situation • Show leadership and confidence regarding the situation • Avoid panic.

Slide

Optimise safety for passengers and protection of property

- Check out the situation the vehicle is in
- Work out a plan
- Do not rush
- Act – rather than react

(Continued)



Slide 106

Slide No	Trainer Notes
106.	Trainer continues to present advice of what driver/tour crew must do when vehicle gets trapped: ● Inspect the situation to determine and gain first-hand knowledge and an understanding of the situation – rather than guess at what it might be ● Work out a plan of action ● Not rush – but take a deep breath and think things through ● Act – rather than react.

Slide

Optimise safety for passengers and protection of property

During a recovery there may be a need to:

- Instruct passengers to leave the vehicle
- Carry items from the vehicle
- Direct passengers to a safe location



Slide 107

Slide No	Trainer Notes
107.	Trainer notes depending on the rescue situation, there may be a need to: ● Remove passenger belongings from the vehicle and/or trailer – and remove to a safer position ● Instruct passengers to leave the vehicle – this may be: ▪ In a sequence determined by the driver ▪ Via a nominated door/exit ● Carry other items from the vehicle – to (for example): ▪ Provide shelter ▪ Provide food and drink to group ▪ Lighten load ● Direct passengers to a safe location (out of 'Danger Zone') – so they: ▪ Can observe proceedings ▪ Are safe – in the event there is a malfunction during the recovery process ▪ Are not located beneath/below the vehicle – in case it rolls or topples ▪ Are not positioned in the path of intended travel of the recovery vehicle, the trapped vehicle and/or the proposed exit route.

Slide

Optimise safety for passengers and protection of property

Regarding the involvement of passengers with the actual vehicle recovery process:

- Follow Tour Operators SOPs in relation to this – some allow it; some do not
- Passengers may see their involvement in the recovery as part of the experience and excitement of the tour



Slide 108

Slide No	Trainer Notes
108.	Trainer discusses involvement of passengers with vehicle recovery noting: • It may be necessary or part of the tour experience to ask a passenger to assist with the recovery of a vehicle • Always follow Tour Operator SOPs in this regard as some operators may prohibit the involvement of passengers in recoveries • In practice, some passengers regard involvement in such activities as part of the fun and experience of their tour.

Slide

Optimise safety for passengers and protection of property

Passengers may be involved in terms of:

- Being spotters and giving directions
- Helping with shovel work
- Assisting with removal of obstacles
- Moving items from and back to the vehicle



Slide 109

Slide No	Trainer Notes
109.	Trainer identifies tasks passengers may be involved in when recovering a vehicle: ● Act as spotters for the recovery – which may relate to: ▪ Giving directions regarding wheel placement and turn of vehicle ▪ Advising of take-up of slack ▪ Notifying of progress using hand signals, or verbal advice including two-way radio ● Help with shovel work ● Assist with removing obstacles ● Aid in moving property to and from the vehicle.

Slide

Inspect vehicle after recovery and perform maintenance/repairs before proceeding

It is vital to understand:

- Greatest potential for repair or maintenance is after a water crossing
- Future safety and performance of the vehicle depends on this type of attention
- Tourist Operators may prescribe required checks
- Many tasks are deferred until vehicle returns at the end of the trip/tour



Slide 110

Slide No	Trainer Notes
110.	Trainer highlights it is important to understand: ● The greatest potential for maintenance and repairs follow water crossings and situations where the vehicle has been immersed in water ● The future safety and performance of the tour vehicle may depend on immediate action being taken after the recovery to 'clear and clean' the vehicle ● Tourism Operators may provide a list of checks to be undertaken – as part of their SOPs or Emergency Management planning ● Many service tasks are not undertaken immediately after the vehicle has been recovered – but may be: ▪ Delayed/deferred until the vehicle returns to base ▪ Undertaken only when an authorised dealer/repair agency is available.

Slide

Inspect vehicle after recovery and perform maintenance/repairs before proceeding

Checks/action may include:

- Checking there is no water in air intake/engine
- Inspecting tyres for damage (tread and sidewalls)
- Re-inflating tyres
- Removing spark plugs or injectors and turning engine over
- Drying out the brakes

(Continued)



Slide 111

Slide No	Trainer Notes
111.	Trainer identifies action/checks may include: ● Checking there is no water in the air intake ● Inspecting tyres – for damage to tread and/or sidewalls ● Re-inflating tyres which have had pressure reduced ● Removing spark plugs, or injectors and turning the engine over to remove water ● Allowing and encouraging brakes to dry out. Classroom Activity – Demonstration and Practical Trainer shows students how to perform tasks listed on the slide and gives them opportunity to practice the same.

Slide

Inspect vehicle after recovery and perform maintenance/repairs before proceeding

- Testing systems and instruments
- Cleaning vehicle glass
- Replacing plugs or injectors
- Applying water displacement products/sprays (WD-40) to electrical parts

(Continued)



Slide 112

Slide No	Trainer Notes
112.	Trainer continues to present action/checks: • Testing operation of systems and instruments • Cleaning vehicle windscreen, mirrors and windows • Replacing plugs, or injectors • Applying water displacement products (such as WD-40) to dry out high and low tension electrical circuits. Classroom Activity – Demonstration and Practical Trainer shows students how to perform tasks listed on the slide and gives them opportunity to practice the same.

Slide

Inspect vehicle after recovery and perform maintenance/repairs before proceeding

- Fixing panel damage
- Noting diffs and transmission needs to be checked after immersion in water (to remove any water which may be present)
- Rolling recovery cables back onto winch spools
- Greasing and oiling parts
- Removing physical debris



Slide 113

Slide No	Trainer Notes
113.	Trainer continues to present action/checks: ● Re-aligning panel damage to enable free movement of wheels and steering ● Making note to have differential and transmission checked (for water) and/or serviced at nearest opportunity ● Rolling cables and ropes back onto spools of winches ● Greasing and oiling nominated parts ● Removing visible debris – from areas and parts such as: ▪ Winches ▪ Winch cables ▪ Radiator ▪ Front of vehicle ▪ Engine bay ▪ Wheel arches ▪ Suspension/under vehicle areas. Classroom Activity – Demonstration and Practical Trainer shows students how to perform tasks listed on the slide and gives them opportunity to practice the same.

Slide

Element 3 – Provide basic maintenance for a 4-WD vehicle

Performance Criteria for this Element are:

- Identify minor faults
- Determine cause of faults
- Conduct visual inspection of and operational tests on vehicle
- Effect necessary repairs
- Effect necessary preventative maintenance



Slide 114

Slide No	Trainer Notes
114.	Trainer identifies for trainees the Performance Criteria for this Element, as listed on the slide. Class Activity – General Discussion Trainer leads a general class discussion by asking questions such as: • What are examples of minor repairs? • What vehicle repairs and maintenance have you undertaken? • What types of vehicle have you worked on? When? Where? • What is preventative maintenance?

Slide

Identify minor faults

Ways to identify vehicle faults:

- Talk with those who have driven or been in the vehicle:
- Asking questions
- Listening to what they say
- Getting them to indicate problems
- Test driving the vehicle

(Continued)



Slide 115

Slide No	Trainer Notes
115.	Trainer presents ways to identify vehicle faults: • Talking with those who have operated/driven or been in a vehicle. Actions required include: ▪ Asking open and closed questions ▪ Asking probing questions ▪ Using terminology appropriate to the individual – avoiding complex terms if the person does not understand them ▪ Listening to what is said ▪ Getting them to point to locations of problems they have identified ▪ The discussion should aim to: – Identify type and nature of fault – Describe the fault if it cannot be identified – Explain whereabouts it is located (if appropriate) – Describe symptoms and indicators – Give an idea of frequency of the problem – Identify conditions under which fault occurs/presents

- | | |
|--|---|
| | <ul style="list-style-type: none">• Test driving the vehicle – to:<ul style="list-style-type: none">▪ Attempt to induce the fault – which can often be a frustrating experience as a problem 'on tour' often fails to manifest itself on return▪ Listen to sounds made by the vehicle▪ Get a feel for the vehicle – and how it handles and responds▪ Smell any odour which may be generated by the fault▪ Duplicate causal conditions which were known to trigger the problem▪ Compare the way it is 'now' to how it normally behaves – in some cases a 'fault' with a vehicle as identified by someone else can simply be a 'lack of performance' which is really not a <i>fault</i> but simply the standard operating parameters of the vehicle. |
|--|---|

Slide

Identify minor faults

- Running the vehicle in a stationary position
- Reading vehicle reports and logs



Slide 116

Slide No	Trainer Notes
116.	Trainer continues to present ways to identify vehicle faults: ● Running stationary vehicle to induce fault – which may involve: ▪ Applying handbrake and/or chocking vehicle ▪ Running vehicle from a cold start ▪ Starting vehicle from a hot start ▪ Running vehicle with nominated accessories turned 'on' (and/or 'off') as required ▪ Running vehicle for nominated period of time ▪ Monitoring/observing vehicle from inside and/or outside of the vehicle ● Reading reports which have been prepared on the vehicle – such as: ▪ Roadworthy/unroadworthy certificates ▪ Vehicle defect notices ▪ Service reports ▪ Daily ('Start of Day') vehicle inspection checklists ▪ End of Tour reports ▪ Damage reports ▪ Maintenance requests ▪ Usage log.

Slide

Determine cause of faults

In reality the process of identifying cause of faults ('diagnosing faults' or 'troubleshooting'):

- Is often not straight-forward
- Can be very time consuming
- Can be difficult in terms of replicating the problem so the fault/cause can be identified
- Can be more difficult than actually fixing the problem



Slide 117

Slide No

Trainer Notes

117.

Trainer provides context stating:

- Identifying the cause of faults may also be known as:
 - 'Diagnosing' faults
 - 'Troubleshooting'
- In many cases the reality of this stage is:
 - It is not straight-forward
 - The process can be very time consuming
 - It is difficult to re-produce the fault so the cause can be found
 - It can be harder to determine the cause than it is to fix the fault.

Classroom Activity – Guest Speaker (or Excursion to garage)

Trainer arranges for qualified mechanic to attend and:

- Explain option for fault identification
- Give hints and tips for identifying faults
- Demonstrate fault diagnoses.

Slide

Determine cause of faults

Practices for diagnosing faults:

- Analysing information gained from fault identification process
- Using diagnostic flowcharts
- Using on-board diagnostics

(Continued)



Slide 118

Slide No	Trainer Notes
118.	Trainer provides context stating: ● Identifying the cause of faults may also be known as: ▪ 'Diagnosing' faults ▪ 'Troubleshooting' ● In many cases the reality of this stage is: ▪ It is not straight-forward ▪ The process can be very time consuming ▪ It is difficult to re-produce the fault so the cause can be found ▪ It can be harder to determine the cause than it is to fix the fault. Classroom Activity – Guest Speaker (or Excursion to garage) Trainer arranges for qualified mechanic to attend and: ● Explain option for fault identification ● Give hints and tips for identifying faults ● Demonstrate fault diagnoses.

Slide

Determine cause of faults

- Referring to service, owner, operator manuals
- Seeking advice from vehicle manufacturer
- Referring to personal experience

(Continued)



Slide 119

Slide No	Trainer Notes
119.	Trainer continues to present practices for diagnosing vehicle faults: ● Referring to service and operator manuals – as most <i>minor</i> faults can be determined with reference to these books ● Seeking additional advice from the manufacturer – which may take the form of: ▪ On-line advice – using: - Free, password protected access to contracted assistance provided as part of new vehicle purchase - Fee-for-service specialist advice ▪ Over the phone assistance – from the manufacturer or the agent/business that sold the vehicle ▪ In-person help – which may include: - Talking with the Service Manager at the agency - Taking the vehicle in to the business for one of the mechanics to look at and give free-of-charge advice on – this is a viable option where the organisation purchases significant vehicles from the one car dealership and/or where a 'preferred supplier' arrangement exists with the business ● Referring to personal experience – the more experience a person has in diagnosing faults, the greater their pool of experience becomes ● This personal experience generates a useful source of information as it is based on 'legitimate', first-hand practice.

Slide

Determine cause of faults

- Loading vehicle onto jacks or stands
- Using sensory appraisal
- Monitoring on-board instrument panel
- Monitoring externally-attached testing equipment



Slide 120

Slide No	Trainer Notes
120.	Trainer continues to present practices for diagnosing vehicle faults: ● Loading vehicle onto jacks, stands or a hydraulic lift – in order to: ▪ Get a better look at the vehicle and be able to see ‘up and under’ it and the engine bay. Certainly, raised vehicles always make it easier to identify the location of leaks and drips ▪ Free up certain suspension components ▪ Make it easier to manipulate other parts of the vehicle ● Using applicable sensory appraisal to help determine cause – this is similar to using the senses to help identify a fault in the first place, and can (depending on the issue) rely on: ▪ Sight – for noting smoke emissions, loose connections and broken and missing parts ▪ Feel – for identifying vibrations ▪ Smell – for detecting unusual odours from the vehicle such as running rich, ‘electrical’ and rubber smells ▪ Hearing – for perceiving exhaust (‘noise emissions’) and engine problems (‘mis-fires’, ‘racing’, ‘knocks’) when the vehicle is running

- | | |
|--|---|
| | <ul style="list-style-type: none">• Monitoring on-board instrument panel – which can mean watching and interpreting the readings of:<ul style="list-style-type: none">▪ Temperature gauges▪ Tachometers▪ Oil pressure▪ Battery charge• Monitoring externally attached equipment – such as:<ul style="list-style-type: none">▪ Various diagnostic tools▪ Battery/alternator testers▪ Vacuum and fuel pump testers▪ Electronic brake service tool▪ Compression test▪ Circuit testers and multi-meters. |
|--|---|

Slide

Conduct visual inspection of and operational tests on vehicle

Activities in the visual inspection process:

- Look at and under the vehicle – lift the vehicle off the ground if possible
- Light the areas being inspected

(Continued)



Slide 121

Slide No	Trainer Notes
121.	Trainer describes activities involved in the inspection process: ● Looking at and under the vehicle and closely checking all aspects of the vehicle – while it is: ▪ On the ground, and/or ▪ On a hoist or over an inspection pit – if a hoist or inspection pit is not available the inspection can occur by using: ▪ Trolley jacks ▪ Car stands ▪ Under-vehicle car trolleys/creepers ● Lighting the areas being inspected – with: ▪ Inspection lights ▪ Torches. Classroom Activity – Demonstration and Practical Trainer demonstrates and explains the vehicle inspection activities shown on the slide and provides opportunity for students to do the same.

Slide

Conduct visual inspection of and operational tests on vehicle

- Use a 'Vehicle Inspection Checklist'
 - Inspect the vehicle in pairs
- (Continued)



Slide 122

Slide No	Trainer Notes
122.	Trainer continues describing activities involved in the inspection process: ● Using a checklist – to: ▪ Provide evidence the inspection occurred ▪ Note defects and matters requiring attention ● Inspecting the vehicle in pairs – so that: ▪ Issues can be analysed/discussed ▪ A more effective inspection takes place – that is, there is less chance something will be missed if two people do the inspection. Classroom Activity – Handout Trainer distributes and discusses use of sample vehicle inspection checklist.

Slide

Conduct visual inspection of and operational tests on vehicle

- Look for signs of damage
- Look for missing parts

(Continued)



Slide 123

Slide No	Trainer Notes
123.	Trainer continues describing activities involved in the inspection process: ● Looking for signs of damage to the vehicle – as these signs can indicate: ▪ Where a more detailed inspection needs to occur to determine: - If the damage <i>has caused a problem</i>/further problem - Whether <i>the damage is a result</i> of another issue ▪ Need for immediate action – to: - Fix the cause - Take temporary action to reduce/prevent further damage - Keep the vehicle on-the-road/operational ▪ Need to take later action – which may not be <i>immediately</i> necessary to ▪ Possible cause or source of the damage ▪ What may be needed/possible to prevent recurrence ● Checking for missing parts – from: ▪ Inside the cabin • Inside the passenger area ▪ On the exterior of the vehicle ▪ Under the hood/bonnet.

	Classroom Activity – Demonstration and Practical
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	Trainer demonstrates and explains the vehicle inspection activities shown on the slide and provides opportunity for students to do the same.
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Slide

Conduct visual inspection of and operational tests on vehicle

- Note leaking fluids
- Identify projections from the vehicle
- Take on-the-spot action to fix simple problems
- Check condition of vehicle components
- Check fluid levels



Slide 124

Slide No	Trainer Notes
124.	Trainer continues describing activities involved in the inspection process: ● Noting leaking fluids – as this can: ▪ Identify type of fluid ▪ Identify location of leak ▪ Indicate severity of problem ▪ Determine, to an extent, the type of problem needing to be addressed ● Identifying projections from the vehicle – this relates to: ▪ Body parts which have been worked loose, damaged or snagged and are hanging loose or sticking out from beyond the normal dimensions of the vehicle presenting a danger to other vehicles, people or structures ▪ Items caught up under the vehicle or under wheel wells, around wheels and brake lines, in the engine bay or from vehicle protection bars or lighting bars ● Taking immediate action to rectify problems – which can be quickly and easily fixed on-the-spot, such as: ▪ Clearing blockages ▪ Tightening screws, nuts, bolts and fittings

- Checking the condition/appearance of critical components of the vehicle (which may involve physical touching, moving, squeezing and removing of parts) such as:
 - Radiator hoses
 - Timing belts
 - Fan belts
 - Tyres including all spares
 - Oil, air and fuel filters
 - Battery/batteries, battery leads, posts, connections, brackets and mountings
- Checking fluid levels – of:
 - Water – radiator and battery/ies
 - Engine oil
 - Transmission fluid
 - Brake fluid
 - Power steering fluid.

Classroom Activity – Demonstration and Practical

Trainer demonstrates and explains the vehicle inspection activities shown on the slide and provides opportunity for students to do the same.

Slide

Conduct visual inspection of and operational tests on vehicle

Basic tests to conduct:

- Operation and condition of lights
- Condition and travel of vehicle pedals
- Operation of parking brake



Slide 125

Slide No	Trainer Notes
125.	Trainer presents basic tests to conduct: ● Lights – in terms of: ▪ Right and left indicator lights – front and rear (including trailers) Headlights – left and right, low beam and high beam; Driving lights and fog lights – on and off; driving lights turn off when low beam selected; Parking lights – front and rear; Tail lights; Reversing lights; Brake lights; Number plate light – front and rear; Side marker lights; Hazard warning lights; Reflectors; Internal vehicle/cabin lights; Instrument lights and signals ▪ Lights should be: - Clean and clear; Without cracks or damage; Correct colour; Firmly and correctly fitted ● Condition of vehicle pedals – for example: ▪ Checking engagement point of brake and clutch pedals ▪ Confirming travel of pedals and freedom from blockages/interference ▪ Activating knobs and switches and confirming they function as required ● Operation of parking brake (also known as the 'hand brake') – to confirm: ▪ Hand brake can be pulled on and will hold the vehicle on a nominated gradient (say, 12% gradient) ▪ Hand brake can be easily and comfortably released to the full 'off' position.

Slide

Effect necessary repairs

To change light globe:

- Identify light to be changed
- Obtain correct globe (or unit)
- Remove light cover
- Unplug electrical connector
- Remove entire unit, if necessary

(Continued)



Slide 126

Slide No	Trainer Notes
126.	Trainer describes how to change light globe in vehicle: ● Identify the light to be replaced ● Obtain correct globe for the job – sealed units will require the entire headlamp to be replaced; semi-sealed units only require the light globe ● Remove the light cover/dust cover ● Unplug electrical connector ● In some cases there is a need to remove the entire light assembly to replace a light globe. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://www.supercheapauto.com.au/how-to/Outside-the-Car/Checking-and-Changing-an-Automotive-Light-Globe.aspx?id=79 – Checking and changing an automotive light globe https://www.youtube.com/watch?v=6hURsa46t4M – How to change a headlight bulb https://www.youtube.com/watch?v=jk1kFc56dwk – How to replace headlight bulbs https://www.youtube.com/watch?v=cB99QN1AYBU – How to install replace front lower side reflector

	https://www.youtube.com/watch?v=mgGWFqPe0Ms – Headlight bulb replacement service: 2005 Honda CRV https://www.youtube.com/watch?v=5F31wzh60zo – How to replace taillight https://www.youtube.com/watch?v=9POigqpLxQc – How to replace fog light and bulb.
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Slide

Effect necessary repairs

- Disconnect globe from holder
- Remove old globe noting how it sits
- Insert new globe – avoid glass contact with fingers
- Re-attach connectors and fittings
- Test



Slide 127

Slide No	Trainer Notes
127.	Trainer continues to describe how to change light globe in vehicle: ● Disconnect whatever mechanism is used to keep the old globe in place ● Remove the old globe noting how it sits – bayonet globes are removed by rotating them anti-clockwise ● Put the new globe in position taking care not to touch the glass part of the globe with the fingers ● Re-attach the mechanism, electrical connection and cover ● Test the light to make sure it is working. Classroom Activity – Demonstration and Practical Trainer shows students how to change a variety of vehicle globes and provides an opportunity for them to do the same

Slide

Effect necessary repairs

To change fan belt:

- Release tension on old belt at bolt on tensioner rail
- Move alternator to allow free play in old belt
- Remove old belt

(Continued)



Slide 128

Slide No	Trainer Notes
128.	Trainer describes how to change fan belt in vehicle: ● Release tension on old belt at bolt on tensioner rail ● Move alternator to allow free play in old belt ● Remove old belt. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=4hnOUs9eiN4 – Change a fan belt http://www.supercheapauto.com.au/how-to/Serviceing-Mechanical/Replacing-a-Drive-Belt.aspx?id=76 – Replacing a drive belt https://www.youtube.com/watch?v=Wg1AX77xEBO – How to replace a serpentine belt https://www.youtube.com/watch?v=01J3Fp9_bMA – How to replace a serpentine belt: Toyota Corolla https://www.youtube.com/watch?v=XD0LNkFYPGQ – Toyota timing belt http://www.wikihow.com/Change-an-Automotive-Belt.

Slide

Effect necessary repairs

- Fit new belt looping it over alternator and water pump pulleys
- Re-locate alternator to bring tension on the belt
- Tighten bolt on tensioner rail
- Run engine and adjust belt if/as required



Slide 129

Slide No	Trainer Notes
129.	Trainer continues to describe how to change fan belt in vehicle: ● Fit new belt by looping it over alternator and water pump pulleys ● Re-locate the alternator to bring tension on the new belt ● Tighten the bolt on the tensioner rail ● Run the engine to test for: ▪ Squealing noise – indicating belt is slipping and needs to be re-tensioned ▪ Dashboard warning lights – which may indicate a problem. Classroom Activity – Demonstration and Practical Trainer shows students how to change a fan belt and provides an opportunity for them to do the same

Slide

Effect necessary repairs

To change/replace a fuse:

- Locate fuse box/block
- Remove cover
- Read key to fuses
- Remove old fuse

(Continued)



Slide 130

Slide No	Trainer Notes
130.	Trainer describes how to change/replace a fuse in vehicle: • Locate fuse box/block • Remove cover • Read key on cover (or in manual) to determine the fuse to replace • Remove fuse to be replaced. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=kOXeo-kqg68 – How to replace a blown fuse in a car https://www.youtube.com/watch?v=faWZmG0fXy8 – How to find and replace a blown fuse in your car or truck.

Slide

Effect necessary repairs

- Insert new fuse
- Verify spare fuses remain available for later use
- Test vehicle to verify new fuse is working
- Replace fuse box/block cover



Slide 131

Slide No	Trainer Notes
131.	Trainer continues to describe how to change/replace a fuse in vehicle: ● Insert new fuse – press in position with finger ● Verify spare fuses are available for emergency use – where space provided to store them ● Test vehicle/accessory/light which was operating on this fuse ▪ If fuse blows immediately then this usually indicates another problem (a short) and the vehicle may need to be taken to an auto-electrician ● If light/accessory functions as required, replace fuse box/block cover. Classroom Activity – Demonstration and Practical Trainer shows students how to replace a fuse and provides an opportunity for them to do the same

Slide

Effect necessary repairs

To change radiator hose:

- Allow motor to cool – remove radiator cap
- Drain radiator – by plug or removing bottom hose
- Use pliers to release clamp holding hose in place
- Replace thermostat at this time too?

(Continued)



Slide 132

Slide No	Trainer Notes
132.	Trainer describes how to change coolant/radiator hose in vehicle: ● Ensure engine has cooled before removing radiator cap ● Drain radiator – if necessary, using drain plug if fitted: there may be a requirement to retain the fluid to put back in the radiator ▪ Remove bottom radiator hose from radiator if there is no drain plug – use ● Use pliers to release clamp holding hose in place – at front and back of top and bottom hose (if changing both hoses) ● It may be a good time to replace the thermostat while the top hose is off the vehicle. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=AGmQ50naiGg – How to replace a radiator hose (upper and lower) https://www.youtube.com/watch?v=C64z0L9816A – Replacing radiator hose https://www.youtube.com/watch?v=xH0D22ITjQU – How to replace the upper radiator hose.

Slide

Effect necessary repairs

- Clean parts to which hose was clamped
- Determine if new hose clamps are needed
- Position clamps near end of hose

(Continued)



Slide 133

Slide No	Trainer Notes
133.	Trainer continues to describe how to change coolant/radiator hose in vehicle: • Clean the parts to which the hose ends were secured – to remove visible debris • Determine if new hose clamps need to be used to fit the new hose/s – use new clamps where indicated • Position clamps (using pliers to open them up) near end of new hoses and replace hoses.

Slide

Effect necessary repairs

- Re-attach hose to radiator tightening clamps
- Fill radiator
- Run and check for leaks

(Continued)



Slide 134

Slide No	Trainer Notes
134.	Trainer continues to describe how to change coolant/radiator hose in vehicle: • Re-attach clamps • Fill radiator • Run and check for leaks. Classroom Activity – Demonstration and Practical Trainer shows students how to change a radiator hose and provides an opportunity for them to do the same

Slide

Effect necessary repairs

Points to note when checking/topping up radiator:

- Allow engine to cool
- Check on level ground
- Add coolant if required
- Wear protective eyewear

(Continued)



Slide 135

Slide No	Trainer Notes
135.	Trainer discusses checking and topping up radiator water stating important points to note: • Engines should be cold when checking and replenishing water in the radiator to avoid safety issues associated with high temperatures and water under pressure • Check fluid levels on level ground • Many radiators require the addition of coolant to the water when replenishing radiator levels – this may be pre-mixed, or require mixing as part of the process. Always follow instructions on the product when adding/mixing coolant • Wear protective eyewear when handling coolant.

Slide

Effect necessary repairs

Topping-up radiator:

- Remove cap from radiator or water reservoir/tank
- Mix coolant and water
- Add mixture observing 'Min-Max' levels
- Replace cap/cover



Slide 136

Slide No	Trainer Notes
136.	Trainer explains topping up radiator may entail: ● Removing cover/cap from engine coolant/water tank or reservoir ● Mixing coolant with water according to manufacturer's recommendations (say, 50:50) if product is not pre-mixed – the preference is to use distilled water for this purpose ● Adding mixed coolant to engine coolant/water tank or reservoir – observe 'Min – Max' marks on the tank/reservoir and do not over-fill ● Replacing and securing cover/cap ● In some situations it may be necessary (or even acceptable) to simply top-off the engine coolant/water tank or reservoir with tap water. Classroom Activity (1) – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=vEdtHZwLKZ8 – Basic car care and maintenance: checking car radiator coolant level Classroom Activity (2) – Demonstration and Practical Trainer shows students how to top-up radiator and provides opportunity for them to do the same.

Slide

Effect necessary repairs

To top-up brake fluid:

- Wear protective gloves and eyewear
- Identify level of brake fluid in brake fluid reservoir
- Clean top and around cap of brake fluid reservoir before removing it

(Continued)



Slide 137

Slide No	Trainer Notes
137.	Trainer explains topping up brake fluid may entail: • Wear protective eyewear and gloves • Identify the level of the brake fluid in the brake fluid reservoir – top-up when level is below the 'Min' mark • Clean the top of, and around, the cap of the brake fluid reservoir prior to opening the unit – to make sure no debris falls in. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=LZ8BZa7d0CQ – How to check and top-up brake fluid: Nulon https://www.youtube.com/watch?v=nJV0J9kTZ2g – Top up your car's brake fluid.

Slide

Effect necessary repairs

- Pour brake fluid directly into brake fluid reservoir
- Fill only to 'Max' line
- Replace and secure cap of brake fluid reservoir
- Wipe away any spills



Slide 138

Slide No	Trainer Notes
138.	Trainer continues to explain topping up brake fluid may entail: ● Pour brake fluid direct from container into the brake fluid reservoir – take care not to spill, and use fresh product where possible ● Fill only to 'Max' line/level on the brake fluid reservoir – do not over-fill ● Replace and secure the cap of the brake fluid reservoir ● Wipe away any spills. Classroom Activity – Demonstration and Practical Trainer shows students how to top-up brake fluid and provides opportunity for them to do the same.

Slide

Effect necessary repairs

Points to note when checking auto transmission fluid:

- Wear protective eyewear
- Park on flat ground
- Make sure engine is at operating temperature
- Check to see if engine needs to be running or not



Slide 139

Slide No	Trainer Notes
139.	Trainer discusses checking automatic transmission fluid noting: • Wear protective eyewear when handling transmission fluid • Park on flat/level ground • Make sure engine is at operating temperature when checking the fluid – not cold • Some makes and models may need to be checked when engine is running and some do not need the engine to be running – follow requirements in Owner Manual.

Slide

Effect necessary repairs

Topping-up automatic transmission fluid:

- Locate correct dipstick
- Remove dipstick
- Wipe dipstick clean
- Re-insert dipstick
- Remove dipstick

(Continued)



Slide 140

Slide No	Trainer Notes
140.	Trainer describes topping-up automatic transmission fluid: • Locate the dipstick for the transmission fluid in the engine bay – ensure the dipstick is not the one for the engine oil • Remove the dipstick • Wipe clean with paper towel or clean rag • Re-insert dipstick – to get reading • Remove dipstick.

Slide

Effect necessary repairs

- Read level on dipstick
- Top-up as required (small amounts) using funnel
- Check level after each filling – avoid over-filling
- Wipe up spills
- Replace and secure dipstick



Slide 141

Slide No	Trainer Notes
141.	Trainer continues to describe topping-up automatic transmission fluid: • Read level of fluid on the dipstick – if level is below 'Min' it needs to be topped-up • Top-up by inserting funnel into dipstick hole and pouring fluid slowly into funnel – base amount added on reading from dipstick but add small amounts • Re-check level after each addition – avoid over-filling • Wipe up any spills • Replace and secure dipstick when 'Normal' level is attained. Classroom Activity – Demonstration and Practical Trainer shows students how to top-up automatic transmission fluid and provides opportunity for them to do the same

Slide

Effect necessary repairs

To check and top-up engine oil:

- Park on level ground
- Check oil with engine 'off' and cold
- Locate dipstick
- Remove dipstick

(Continued)



Slide 142

Slide No	Trainer Notes
142.	Trainer describes checking and topping-up engine oil: • Park on flat/level ground • Check the engine oil with motor turned off and cold – do not check when engine is running and/or when it is hot • Locate the dipstick for the engine oil in the engine bay • Remove the dipstick. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=6lQBZm8DBAw – Checking and adding oil to your vehicle for beginners http://m.supercheapauto.com.au/How-To/Oils-Lubricants/Checking-Topping-Up-Engine-Oil/194 - Checking and topping up engine oil

Slide

Effect necessary repairs

- Wipe dipstick clean
- Re-insert dipstick
- Remove dipstick
- Read dipstick
- Locate and clean around filler cover

(Continued)



Slide 143

Slide No	Trainer Notes
143.	Trainer continues to describe checking and topping-up engine oil: ● Wipe clean with paper towel or clean rag ● Re-insert dipstick – to get reading ● Remove dipstick ● Read level of oil on the dipstick – if level is below 'Min' it needs to be topped-up ● Locate oil filler cap on top of motor – clean around the cap before removing it. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=uvUPsk2bN0A – Check your fluids, engine oil, transmission fluid, power steering, coolant etc. https://www.youtube.com/watch?v=BSlouL8Ekgw – How to check car fluids, belts and perform full underhood inspection

Slide

Effect necessary repairs

- Use funnel to pour oil into engine
- Add small amounts each time
- Only use oil recommended by manufacturer

(Continued)



Slide 144

Slide No	Trainer Notes
144.	Trainer continues to describe checking and topping-up engine oil: • Top-up oil by inserting funnel into hole and pouring fluid slowly into funnel – base amount added on reading from dipstick • Add small amounts each time • Only use oil recommended by vehicle manufacturer.

Slide

Effect necessary repairs

- Check level after each addition
- Avoid over-filling
- Clean spills
- Replace and secure dipstick



Slide 145

Slide No	Trainer Notes
145.	Trainer continues to describe checking and topping-up engine oil: • Re-check level after each addition • Do not overfill • Clean away any spills/drips • Replace and secure dipstick and oil filler cap when required level is attained – between 'Min' and 'Max' indicators on the dipstick. Classroom Activity – Demonstration and Practical Trainer shows students how to top-up engine oil and provides opportunity for them to do the same

Slide

Effect necessary repairs

SOP if a tour vehicle gets a puncture:

- Remove and replace with spare
- Call in to a repair centre to have puncture repaired
- Do not repair tyre with damaged sidewall
- Do not undertake own repair of puncture unless it is necessary to do so



Slide 146

Slide No	Trainer Notes
146.	Trainer discusses addressing punctures to a tour vehicle stating if a tour vehicle suffers a puncture while on tour the standard responses are to: ● Remove the punctured tyre and replace with one of the spares carried on the vehicle ● Call in to a repair centre, tyre agency or garage as soon as possible to have the punctured tyre repaired ● Never try to repair a tyre where the sidewall is damaged ● The standard response to a puncture is, therefore, not to repair the puncture until it becomes absolutely necessary to do so.

Slide

Effect necessary repairs

Options for repairing punctures:

- Use a pressurised can of tyre adhesive
- Use a tyre repair kit to plug the hole



Slide 147

Slide No	Trainer Notes
147.	Trainer identifies and explains two options for puncture repair: ● Use a pressurised commercial can of tyre adhesive to provide a short-term solution: ▪ Following the instructions on the can ▪ Reducing driving speed until a professional repair can be made ▪ Getting the tyre repaired professionally at the earliest opportunity ● Use a tyre repair kit to plug the hole following the instructions which come with the kit – for example: ▪ Remove the cause of the puncture ▪ Lubricate reaming tool and ream out hole with the tool in the kit – to make the hole big enough to take a the plug ▪ Lubricate the plug with some rubber adhesive – and fix into the insertion tool ▪ Use the insertion tool to locate the plug into the hole- - and withdraw the insertion tool ▪ Cut off excess plug protruding from the tread of the tyre ▪ Inflate the tyre as required.

Classroom Activity (1) – Internet research

Trainer supervises internet research on relevant websites such as:

<https://www.youtube.com/watch?v=ulQmDaqv4F4> – Best tire repair video

<https://www.youtube.com/watch?v=LRQJgTxRhOA> – How to repair a puncture in a tyre with a repair plug/string

<http://www.tough toys.com.au/4wd-recovery/tyre-repair-kits-and-equipment/> - Tyre repair kits and equipment.

Classroom Activity (2) – Demonstration and Practical

Trainer shows students how to plug a hole in a tyre using a tyre plug kit and provides opportunity for them to do the same.

Slide

Effect necessary repairs

Bleeding brakes:

- May need to occur if they become soft/spongy
- Can be a two-person job
- All four wheels have to be bled
- Brake fluid is corrosive



Slide 148

Slide No	Trainer Notes
148.	Trainer discusses bleeding brakes observing: ● If brakes become spongy, soft or ineffective on tour they may require bleeding to remove air in the brake lines ● This can be a two-person job – one person sitting in the vehicle and pumping the brake pedal and the other handling the brake bleeder bottle and the spanner to open and close the bleeder nipple/valve ● When bleeding the brakes, all four wheels have to be bled: common practice is to start the process at the wheel furthest from the master cylinder and work through all four wheels finishing with the wheel closest to the master cylinder ● As brake fluid is corrosive this means it is dangerous so appropriate protective gloves and eyewear should be used. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=IPwM2vn3bT0 – How to bleed disc brakes https://www.youtube.com/watch?v=lkd7iq-dlQ0 – Bleeding the brake system.

Slide

Effect necessary repairs

To bleed the brakes:

- Check brake fluid reservoir to ensure sufficient fluid
- Monitor fluid level throughout procedure
- Have replacement fluid on hand
- Jack the vehicle up and remove wheels (optional)
- Attach brake bleeder bottle to brake nipple

(Continued)



Slide 149

Slide No	Trainer Notes
149.	Trainer describes how to bleed the brakes: • Check the brake fluid reservoir – to make sure there is sufficient brake fluid available in the reservoir before starting the process • Make sure the level is monitored throughout the process so air is not re-introduced into the brake system • Have replacement/new brake fluid on hand – to top-up the reservoir as required • Jack the vehicle up and remove the wheels – optional in some cases • Attach brake bleeder bottle to brake nipple.

Slide

Effect necessary repairs

- Open valve and depress brake 4 – 5 times
- Close valve and tighten – top up brake fluid reservoir
- Check for leaks by pressing brake pedal hard
- Remove bleeder bottle – discard fluid
- Repeat for other 3 wheels



Slide 150

Slide No	Trainer Notes
150.	Trainer continues to describe how to bleed the brakes: ● Open valve and depress brake pedal four to five times ● Close valve and tighten – top up reservoir with brake fluid ● Check for leaks by pressing brake pedal hard – re-tighten valve/nipple if there is a leak ● Remove brake bleeder bottle – discard fluid ● Repeat for remaining three wheels – monitoring level of brake fluid in brake fluid reservoir and topping-up as necessary. Classroom Activity – Demonstration and Practical Trainer shows students how to bleed the brakes and provides opportunity for them to do the same.

Slide

Effect necessary repairs

Bleeding a diesel engine:

- Refers to removing air from the fuel lines
- Air locks will prevent the engine starting or running smoothly
- Always follow directions in Owner manual
- Modern vehicles may feature easy-to-use 'bleed screws'



Slide 151

Slide No	Trainer Notes
151.	Trainer discusses bleeding an engine stating: ● Bleeding a diesel engine refers to removing air from the lines so the engine will re-start after, for example, the engine has run out of fuel or there has been a need to replace the fuel filter ● If the engine/fuel lines are not bled, the air in the lines blocks the flow of fuel and the engine will not start (or will run roughly) ● Refer to the Owner's manual/Operator's manual for the vehicle to obtain manufacturer's instructions in this regard as there are different requirements and vehicle components for different makes and models of vehicles and engines ● More modern engines feature alternate (more simple-to-use) systems such as bleed screws whereby the screw is opened, the engine is cranked and then the screw is closed.

Slide

Effect necessary repairs

To bleed an engine:

- Fill with fuel
- Confirm new fuel filter is correctly fitted
- Prime the lift pump
- Open line to injector furthest from pump and allow air to escape
- Tighten line when steady flow of diesel appears
- Clean up spilled fuel
- Repeat for each cylinder



Slide 152

Slide No	Trainer Notes
152.	Trainer describes actions to bleed an engine: ● Filling the vehicle with fuel – to prevent recurrence of the problem if the problem was caused by running out of fuel ● Confirming the new fuel filter is properly fitted – and was filled with fuel prior to being re-fitted ● Prime the lift pump – by activating the lever attached to move fuel to the injector ● Open the line to the injector furthest away from the pump – and allow air to escape ● Tighten the line when there is a steady flow of diesel – and clean away spilt fuel ● Repeat for each cylinder. Classroom Activity – Demonstration and Practical Trainer shows students how to bleed an engine and provides opportunity for them to do the same.

Slide

Effect necessary repairs

To change spark plugs:

- Disconnect HT lead to plug
- Loosen plug with spark plug socket
- Use fingers to remove spark plug
- Take care to not let debris fall into cylinder

(Continued)



Slide 153

Slide No	Trainer Notes
153.	Trainer describes process for replacing spark plugs: • Disconnect the HT lead to the spark plug • Use a spark plug socket attached to a wrench in an anti-clockwise direction to loosen the used spark plug – many plugs can be hard to shift so a long handle will provide the necessary extra pressure • Use the fingers to unscrew the old spark plug and remove it once it has been freed • Take care not to allow any debris to fall into the cylinder.

Slide

Effect necessary repairs

- Clean or obtain new plug
- Use feeler gauge to set correct gap
- Use fingers to re-seat plug into block

(Continued)



Slide 154

Slide No	Trainer Notes
154.	Trainer continues to describe process for replacing spark plugs: ● Clean the plug with a wire brush and check the condition of the plug to determine: ▪ If a new plug is required ▪ If there is evidence of another problem which should be investigated ● Use a feeler gauge to set the correct gap for the old or new spark plug to be fitted ● Use fingers to re-seat the plug into the block. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://www.supercheapauto.com.au/how-to/Service-Mechanical/Replacing-Spark-Plugs.aspx?id=68 – Replacing spark plugs https://www.youtube.com/watch?v=jj06A6UdHvI – Changing spark plugs and replacing plug wires.

Slide

Effect necessary repairs

- Tighten finger tight
- Secure with socket to torque specification
- Re-attach HT lead
- Start and run engine to test



Slide 155

Slide No	Trainer Notes
155.	Trainer continues to describe process for replacing spark plugs: • Tighten gently until finger tight and ensure thread is not crossed • Finish tightening with spark plug socket attached to a torque wrench to required torque specification • Re-attach HT lead • Start and run motor to test. Classroom Activity – Demonstration and Practical Trainer shows students how to change spark plugs and provides opportunity for them to do the same.

Slide

Effect necessary repairs

To replace glow plugs:

- Remove visible debris around plugs
- Disconnect battery
- Use spanner to remove nuts holding bridging strap to plugs
- Remove bridging strap

(Continued)



Slide 156

Slide No	Trainer Notes
156.	Trainer describes process for replacing glow plugs: ● Remove any visible debris around the plugs which may fall into the cylinders – use of compressed air is recommended ● Disconnect the (negative) battery terminal ● Use a spanner to remove the nuts holding the bridging strap to the plugs ● Remove the bridging strap. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=4Po_1pNHE1Y – Easy and complete glow plug test. http://autorepair.about.com/od/engine-relateddiyjobs//aa081404a.htm - How to replace your diesel glow plugs.

Slide

Effect necessary repairs

- Use spanner to loosen plugs and remove with fingers
- Test new plug to be fitted
- Ensure plug and thread is clean
- Screw new plug in by hand until seated in cylinder head

(Continued)



Slide 157

Slide No	Trainer Notes
157.	Trainer continues to describe process for replacing glow plugs: • Use a spanner to untighten the glow plug and remove with fingers • Test the new glow plug to be fitted • Ensure plug and thread is clean • Screw new plug in by hand until seated in cylinder head. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: https://www.youtube.com/watch?v=RCcmN-r5Z9M – Replacing glow plugs in a Mitsubishi Delica L400 https://www.youtube.com/watch?v=4Po_1pNHE1Y – Glow plug change

Slide

Effect necessary repairs

- Tighten with torque wrench to required specification
- Replace bridging strap and nuts
- Re-connect battery
- Test



Slide 158

Slide No	Trainer Notes
158.	Trainer continues to describe process for replacing glow plugs: • Tighten with torque wrench to torque specification according to the individual plug • Replace bridging strap securing with nuts • Re-connect battery • Test. Classroom Activity – Demonstration and Practical Trainer shows students how to change glow plugs and provides opportunity for them to do the same.

Slide

Effect necessary preventative maintenance

Provision of regular service and maintenance to tour vehicles will/can:

- Help extend working life of the vehicle
- Increase reliability of the vehicle
- Reduce operating costs
- Present a better-looking vehicle to tour groups



Slide 159

Slide No	Trainer Notes
159.	Trainer highlights provision of regular service and maintenance to tour vehicles will/can: • Help extend working life of the vehicle • Increase reliability of the vehicle • Reduce operating costs • Present a better-looking vehicle to tour groups – which increases confidence in the tour staff and tour operator.

Slide

Effect necessary preventative maintenance

Ways to determine preventative maintenance requirements:

- Take into account distance travelled and hours vehicle has been used
- Conduct and note response to pre-start checks
- Comply with internal organisational SOPs

(Continued)



Slide 160

Slide No	Trainer Notes
160.	Trainer presents ways to determine preventative maintenance requirements: • Take into account distance travelled and number of hours the vehicle has been used – there is a direct relationship between extensive/long use' and the need to provide preventative maintenance • Conduct and note response to pre-start checks – as designated by vehicle manufacturers specific to individual makes and models of 4WDs which may indicate need for maintenance. These checks commonly involving checks of: ▪ Fuel, water and oil; Brake and transmission fluid levels; Battery water levels and electrolyte checks; Tyres; Belts; Leads; Hydraulic lines and connections; Air cleaners; Air-conditioners; Brakes • Comply with internal organisational SOPs – many Tour Operators will provide a 'Start of day' checklist for each tour vehicle which needs to be followed before the vehicle is used every day. These lists: ▪ Differ between vehicle types to reflect their individual need ▪ Provide a record of the inspection to prove the check was undertaken as required ▪ Record work done as part of the daily inspection ▪ Note where items may require further attention at a later stage/at conclusion of the tour.

Slide

Effect necessary preventative maintenance

- Factor in the nature of roads/tracks travelled
- Use personal observation
- Refer to personal experience
- Research/read manufacturer requirements



Slide 161

Slide No	Trainer Notes
161.	Trainer continues to present ways to determine preventative maintenance requirements: ● Factor in the nature of roads/tracks travelled – some surfaces are more punishing than others and will impose a more urgent need for attention to the vehicle ▪ Surfaces where there has been lots of sand, rocks, water crossings and/or mud are all indicators for some extra inspection/maintenance ● Use personal observation – in terms of: ▪ The way the vehicle is performing/responding ▪ Sensory awareness of changes in noises, vibration and/or smells from the vehicle ▪ Noticing damaged items on the vehicle ▪ Noting alerts from on-board warning lights or instruments ● Refer to personal experience – related to previous incidents which have been experienced with similar vehicles under similar conditions ● Research/read manufacturer requirements in relation to routine servicing and preventative maintenance – in: ▪ Service manuals ▪ Owner manuals ▪ Repair manuals.

Slide

Effect necessary preventative maintenance

Examples of preventative maintenance which may be required for on-tour vehicles:

- Cleaning
- Undertaking scheduled service tasks as recommended by the manufacturer

(Continued)



Slide 162

Slide No	Trainer Notes
162.	Trainer indicates possible preventative maintenance required for on-tour vehicles: ● Cleaning the vehicle – the value of cleaning on tour must never be under-estimated. Cleaning in this context involves: ▪ Removing dirt, sand, mud and grit and grime; Cleaning the engine bay; Cleaning out under the wheel arches; Cleaning the winch; Cleaning lights and reflectors; Cleaning the windscreen and windows ● Undertaking scheduled service tasks as recommended by the manufacturer – tour vehicles will generally be fully serviced prior to departing on tours but there is always the potential for 'scheduled servicing' to have to be provided while on tour ▪ There may be a preference for this service to be undertaken along the route at a qualified service provider but this is not always possible ▪ Where it is not possible, the requirements of the Owner's manual for the specific time period/distance travelled need to be accommodated to the best extent possible ▪ Indicative tasks include: - Replacing belts; Replacing plugs; Draining and replacing fluids and oils; Replacing filters.

Slide

Effect necessary preventative maintenance

- Undertaking specified tasks as nominated by Tour Operator
- Replacing specific component parts
- Examining tyres

(Continued)



Slide 163

Slide No	Trainer Notes
163.	Trainer continues to indicate possible preventative maintenance required for on-tour vehicles: ● Undertaking specified additional service tasks as required by the host enterprise – these can be prescribed by the Tour Operator on the basis of previous experience with vehicles and routes. Normally these will require: ▪ The same attention being paid to the same parts as for a scheduled service but with greater frequency ▪ Attention to specific parts of the vehicle as a result of activities undertaken ● Replacing specific component parts – according to a nominated schedule ● Examining the tyres – and checking and/or adjusting the tyre pressures. Classroom Activity – Internet research Trainer supervises internet research on relevant websites such as: http://www.offroadaussie.com/2013/02/4wd-tips-and-tricks-water-crossings/ - 4WD tricks and tips: water crossing.

Slide

Effect necessary preventative maintenance

- Topping-up fluids
- Tightening loose connections and fittings
- Replacing damaged/worn parts



Slide 164

Slide No	Trainer Notes
164.	Trainer continues to indicate possible preventative maintenance required for on-tour vehicles: ● Topping up of fluids – as required based on inspection ● Tightening loose connections and fittings ● Replacing damaged or worn parts. Classroom Activity – Excursion Trainer arranges visit to local Tour Operator to: ● Talk to operators regarding operation and maintenance of 4WD vehicles for use on tours ● View preventative maintenance being delivered to tour vehicles ● Obtain supporting documentation relating to operation and maintenance of tour 4WD vehicles.

Slide

Effect necessary preventative maintenance

Most tour vehicles need to be prepared in advance of tours – as:

- Dictated by Tour Operator SOPs
- Determined by driver/tour crew – with reference to known itinerary



Slide 165

Slide No	Trainer Notes
165.	Trainer continues to indicate possible preventative maintenance required for on-tour vehicles: • Topping up of fluids – as required based on inspection • Tightening loose connections and fittings • Replacing damaged or worn parts. Classroom Activity – Excursion Trainer arranges visit to local Tour Operator to: • Talk to operators regarding operation and maintenance of 4WD vehicles for use on tours • View preventative maintenance being delivered to tour vehicles • Obtain supporting documentation relating to operation and maintenance of tour 4WD vehicles.

Slide

Element 4 – Report on 4-wheel drive operation

Performance Criteria for this Element are:

- Complete documentation to record services and repair undertaken
- Notify management of issues that require external attention



Slide 166

Slide No	Trainer Notes
166.	Trainer identifies for trainees the Performance Criteria for this Element, as listed on the slide. Class Activity – General Discussion Trainer leads a general class discussion by asking questions such as: • What documentation has to be completed? • Who completes it? • When and where? • Why does it need to be completed?

Slide

Complete documentation to record services and repair undertaken

Reasons documentation needs to be completed:

- Provide proof of daily vehicle inspections
- Give a reference point for work performed
- Allow accurate/comprehensive vehicle records to be kept
- Enable accurate costings

(Continued)



Slide 167

Slide No	Trainer Notes
167.	Trainer discusses need for Tour Operators to have documentation completed to record services and repairs undertaken to tour vehicles when on-tour: • Provide proof of inspections performed on vehicles while on tour • Give a reference point for work which has to be performed on tour vehicles • Allow comprehensive and accurate operational records to be maintained for each vehicle • Enable accurate costing to be undertaken relating to the cost of operation of individual vehicles and types of vehicles.

Slide

Complete documentation to record services and repair undertaken

- Supply service history when vehicle is sold
- Help determine future repairs/services
- Demonstrate compliance with service requirements
- Allow reconciliation of receipts, invoices and related paperwork



Slide 168

Slide No	Trainer Notes
168.	Trainer continues to identify need to complete documentation: ● Supply a database of information which can be provided to buyers when the tour vehicle is on-sold at the end of its useful life – to assist with proving regular maintenance and optimise re-sale value ● Help determine future repairs and maintenance which need to be provided to individual vehicles – based on the work they have done, and any service/maintenance performed on them ● Demonstrate tour crew/vehicle drivers' compliance with internal requirements – regarding daily inspections and scheduled/required maintenance and service ● Allow administration/management of the Tour Operator to reconcile receipts and invoices from external providers for vehicle expenses.

Slide

Complete documentation to record services and repair undertaken

A wallet or folder:

- Is usually provided in each vehicle for holding relevant documents
- May also contain the fuel card for the vehicle



Slide 169

Slide No	Trainer Notes
169.	Trainer explains: • A wallet or folder is usually provided in all vehicles for the purposes of storing required documentation to facilitate the submission and processing of paperwork at the end of the tour • This may contain the fuel card for the vehicle.

Slide

Complete documentation to record services and repair undertaken

Keys when documenting services and repairs:

- Write clearly
- Be honest
- Cover all work done
- Use recognised abbreviations and acronyms
- Complete documentation 'at the time' work is done



Slide 170

Slide No	Trainer Notes
170.	Trainer highlights keys in the documentation process are: • Write clearly • Be honest • Cover all work done – be comprehensive • Use only recognised abbreviations and acronyms • Complete required documentation 'at the time' – as opposed to relying on memory to complete it at a later time.

Slide

Complete documentation to record services and repair undertaken

Daily Vehicle Inspection Checklist
– is completed at start of each day
for each on-tour vehicle and is
used to:

- Record the inspection and prove it took place
- Record work done as part of the check
- Note items which require further/later attention



Slide 171

Slide No	Trainer Notes
171.	Trainer discusses Daily Vehicle Inspection Checklists stating: • These are the 'Start of Day' lists to be completed for each vehicle on every day of the tour before the vehicle is used. The checklist is used to: ▪ Provide a record of the inspection to prove the check was undertaken as required ▪ Record work done (service, repairs and maintenance) as part of the daily inspection ▪ Note where items may require further attention at a later stage/at conclusion of the tour. Classroom Activity – Handout Trainer distributes and discusses sample Daily Vehicle Inspection Checklist.

Slide

Complete documentation to record services and repair undertaken

Activities in completing a Daily Vehicle Inspection Checklist:

- A 'walk around' of the vehicle
- Physical inspection of the vehicle
- Starting engine and running the vehicle

(Continued)



Slide 172

Slide No	Trainer Notes
172.	Trainer explains activities inherent in completing a Daily Vehicle Inspection Checklist: • A walk-around of the vehicle • Physical inspection of the vehicle • Starting of the engine and running the vehicle.

Slide

Complete documentation to record services and repair undertaken

- Monitoring gauges/instrument panel
- Checking leads and connections
- Testing operation of systems and equipment



Slide 173

Slide No	Trainer Notes
173.	Trainer continues to identify activities involved in completing a Daily Vehicle Inspection Checklist: ● Monitoring gauges/instruments ● Checking leads and connections ● Testing the operation of certain systems and/or items of equipment. Classroom Activity – Demonstration and Practical Trainer shows students how to complete a Daily Vehicle Inspection Checklist and provides opportunity for them to do the same.

Slide

Complete documentation to record services and repair undertaken

Vehicle Log/Vehicle Usage Log:

- Records basic details of vehicle use on a daily basis
- Usually has to be completed:
 - At start of day
 - At end of day



Slide 174

Slide No	Trainer Notes
174.	Trainer explains: ● The Vehicle Log/Vehicle Usage Log records basic travel information for each vehicle on a day-by-day basis ● It commonly has to be filled in: ▪ At the start of the day ▪ At the end of the day. Classroom Activity – Handout Trainer distributes and discusses sample Vehicle Log/Vehicle Usage Log .

Slide

Complete documentation to record services and repair undertaken

Information required in Vehicle Log/Vehicle Usage Log:

- Name of tour
- Vehicle details/identification
- Odometer reading – start and end of day
- Distance travelled

(Continued)



Slide 175

Slide No	Trainer Notes
175.	Trainer describes information required when completing a Vehicle Log/Vehicle Usage Log: • Name/title of tour • Details of vehicle – usually by registration number, make and model • Odometer reading/s – start and end of day • Distance travelled – calculated from odometer readings.

Slide

Complete documentation to record services and repair undertaken

- Date/s
- Destination/s and route/s
- Driver name and signature
- Travel time/s



Slide 176

Slide No	Trainer Notes
176.	Trainer describes information required when completing a Vehicle Log/Vehicle Usage Log: • Name/title of tour • Details of vehicle – usually by registration number, make and model • Odometer reading/s – start and end of day • Distance travelled – calculated from odometer readings.

Slide

Complete documentation to record services and repair undertaken

Receipts and invoices:

- Need to be kept and forwarded at end of tour
- Show services provided by third party providers – such as repairs, maintenance, service
- Identify items supplied by third party providers – such as fuel, oil, tyres



Slide 177

Slide No	Trainer Notes
177.	Trainer describes information required when completing a Vehicle Log/Vehicle Usage Log: • Name/title of tour • Details of vehicle – usually by registration number, make and model • Odometer reading/s – start and end of day • Distance travelled – calculated from odometer readings.

Slide

Notify management of issues that require external attention

Reasons to report issues requiring attention – so management can:

- Decide if they want to have the matter attended to or not
- Obtain quotations
- Make sure work to be done is best/most efficiently integrated into known tour schedule

(Continued)



Slide 178

Slide No	Trainer Notes
178.	Trainer describes information required when completing a Vehicle Log/Vehicle Usage Log: • Name/title of tour • Details of vehicle – usually by registration number, make and model • Odometer reading/s – start and end of day • Distance travelled – calculated from odometer readings.

Slide

Notify management of issues that require external attention

- Make arrangements for a replacement vehicle
- Ensure all tour vehicles are maintained in a safe and reliable condition
- Factor this knowledge into other tour-related planning decisions



Slide 179

Slide No	Trainer Notes
179.	Trainer describes information required when completing a Vehicle Log/Vehicle Usage Log: • Name/title of tour • Details of vehicle – usually by registration number, make and model • Odometer reading/s – start and end of day • Distance travelled – calculated from odometer readings.

Slide

Notify management of issues that require external attention

Communication to management while on-tour of tour vehicle issues which will require external attention usually relate to:

- Major repairs
- Significant other work/expenses



Slide 180

Slide No	Trainer Notes
180.	Trainer advises on-tour notification of issues requiring external attention usually relates to: • Major repairs (which are usually defined in monetary terms, such as 'work in excess of X value) – must be notified prior to getting work done so that: ▪ Authorisation to proceed can be obtained – because the Tour Operator may elect not to proceed with repairs ▪ Arrangements for payment can be made – between the Tour Operator and the service provider • Significant other work/expenses which have been incurred need to be communicated on the day they were incurred – as part of the standard <i>daily</i> check in/tour report that the Tour Leader may be required to make, so the Tour Operator is kept informed of issues and progress.

Slide

Notify management of issues that require external attention

On-tour notification may occur:

- Via two-way radio
- Using cell phone
- Via a mobile device



Slide 181

Slide No	Trainer Notes
181.	Trainer advises on-tour notification/reports may be made: • Via two-way radio • Using cell phone • Via a mobile device.

Slide

Notify management of issues that require external attention

Report/notification may also occur at end of tour:

- By providing a verbal report
- By completing a vehicle maintenance requisition
- By discussing the problem/issue with specialist



Slide 182

Slide No	Trainer Notes
182.	Trainer advises notification may also occur when the tour concludes and there can be a need for the vehicle driver to: • Provide a comprehensive verbal report on relevant issues – as part of the Tour de-briefing/end-of-tour report, supplemented by and with reference to relevant documentation (if any) from service providers gathered throughout the tour • Complete a vehicle maintenance requisition – specifying: ▪ Identification of vehicle to be maintained ▪ Nature of the problem ▪ Cause of the problem, if known ▪ Work already done • Suggestions for maintenance made by professional third party providers <i>en route</i> ▪ Discussing the problem with specialists – which is best done: – Face-to-face – While the vehicle is present – so specific points/parts can be indicated and problems replicated in front of mechanics/service providers.

Recommended training equipment

- Vehicles to operate and practice maintenance and minor repairs – selection should embrace:
 - Four-wheel drives
 - Petrol engines
 - Diesel motors
- Range of large and small hand and electrical tools and specialist equipment/tools required to perform maintenance and undertake repairs:
 - Brake bleeder bottle
 - Wrenches – various sizes, imperial and metric
 - Torque wrench
 - Adjustable, open end, box end (6 and 12 sided) wrenches
 - Combination wrenches
 - Ratchets – various sizes, imperial and metric
 - Standard sockets – various sizes, imperial and metric
 - Spark plug sockets
 - Impact sockets
 - Pliers
 - Slip joint pliers
 - Locking pliers
 - Groove joint pliers
 - Needle nose pliers
 - Diagonal cutting pliers
 - Screwdrivers – range of sizes and lengths, flat-head and Phillips head
 - Hammers – ball peen, rubber
 - Pry bars – different sizes/lengths
 - Slide hammer bars
 - Vehicle hoist
 - Wheel jacks – different types: scissor, bottle, pneumatic; trolley
 - Vehicle ramps
 - Wheel chocks
 - Jack stands
 - Tyre pressure gauge
 - Tread depth gauge
 - Four-way tyre tool

- Measuring rulers and tapes – metric and imperial
- Coolant tester
- Sparkplug gauge
- Feeler gauge
- Calipers
- Micrometers
- Dial indicators
- Multi-meters
- Fuse pullers
- Wire strippers
- Battery load tester
- Jumper cables
- Battery terminal puller
- Battery brush
- Battery terminal spreader
- Oil drain pan
- Oil filter wrench
- Funnel – different sizes/shapes
- Grease guns
- Hacksaw
- Grinder
- Chisels
- Punches
- Files
- Hex keys/Allen keys
- Compressors – for driving pneumatic equipment and for tyre inflation
- Vice
- Range of replacement parts/spare parts:
 - Spark plugs
 - Glow plugs
 - Range of nuts, bolts, screws, connectors, electrical wire, fuses
 - HT leads
 - Fan belts and other engine drive belts
 - Radiator hoses – top and bottom
 - Oil filters
 - Fuel filters

- Tyres and tubes
- Tyre deflators
- Tyre cores and caps
- Tyre repair kits
- Range of globes
- Mirrors
- Reflectors
- Windscreen wiper blades
- Fuel – petrol, diesel, LPG
- Range of vehicle recovery gear - such as:
 - Shackles
 - Drag chains
 - Snatch straps
 - Power (vehicle mounted) winches
 - Snatch blocks
 - Hand-operated winch
 - Pulley blocks
 - Extension straps
 - Tree protectors
 - High-lift jacks
 - Under-vehicle inflatable bags
 - Long-handled shovel
 - Portable tracks/sand ladders
 - Extensions
 - Gloves
 - Ground anchors
 - Recovery blankets/dampers
- Range of grease, engine oils, brake fluids, transmission fluids, power steering fluid, coolant, anti-freeze
- EPIRB (Emergency Position Indicating Radio Beacon)
- GPS (Global Positioning System)
- Two-way radios
- Public address systems
- Samples/examples of:
 - Warranties and guarantees
 - Service manuals

- Owner manuals
- Workshop manuals
- Repair manuals
- Documentation completed when a vehicle has been services, repaired and/or maintained
- Vehicle inspection checklists
- 'Start of Day' vehicle inspection checklists
- Service reports/logs/forms
- Safety clothing and equipment (PPE) required for undertaking vehicle repairs/service
- Copies of host country legislation that applies to the maintenance of commercial vehicles used for passenger transport
- It can be useful to watch <https://www.youtube.com/watch?v=0b7XfJJ6Kcl> – Tools of the small engine mechanic.

Instructions for Trainers for using PowerPoint – Presenter View

Connect your laptop or computer to your projector equipment as per manufacturers' instructions.

In PowerPoint, on the **Slide Show** menu, click **Setup Show**.

Under Multiple monitors, select the Show Presenter View check box.

In the **Display slide show** on list, click the monitor you want the slide show presentation to appear on.

Source: <http://office.microsoft.com>

Note:

In Presenter View:

You see your notes and have full control of the presentation

Your trainees only see the slide projected on to the screen

More Information

You can obtain more information on how to use PowerPoint from the Microsoft Online Help Centre, available at:

<http://office.microsoft.com/training/training.aspx?AssetID=RC011298761033>

Note Regarding Currency of URLs

Please note that where references have been made to URLs in these training resources trainers will need to verify that the resource or document referred to is still current on the internet. Trainers should endeavour, where possible, to source similar alternative examples of material where it is found that either the website or the document in question is no longer available online.

Appendix – ASEAN acronyms

AADCP	ASEAN – Australia Development Cooperation Program
ACCSTP	ASEAN Common Competency Standards for Tourism Professionals
AEC	ASEAN Economic Community
AMS	ASEAN Member States
ASEAN	Association of Southeast Asian Nations
ASEC	ASEAN Secretariat
ATM	ASEAN Tourism Ministers
ATPMC	ASEAN Tourism Professionals Monitoring Committee
ATPRS	ASEAN Tourism Professional Registration System
ATFTMD	ASEAN Task Force on Tourism Manpower Development
CATC	Common ASEAN Tourism Curriculum
MRA	Mutual Recognition Arrangement
MTCO	Mekong Tourism Coordinating office
NTO	National Tourism Organisation
NTPB	National Tourism Professional Board
RQFSRS	Regional Qualifications Framework and Skills Recognition System
TPCB	Tourism Professional Certification Board

